MogaHSI: Multiorder Gated Aggregation Network for Hyperspectral Image Classification

Garas Gendy and Hasan AlMarzouqi, *Senior Member, IEEE*

***Abstract*—Hyperspectral imaging (HSI) classification is an essential task in remote sensing applications, necessitating precise modeling of intricate spectral–spatial dependencies. Recently, transformer-based methods have received considerable attention due to their ability to capture global contextual information. However, in the HSI domain, they face additional challenges beyond quadratic computational complexity. The high spectral redundancy in hyperspectral data makes token representations less efficient and increases memory and computational overhead, making them difficult to use in resource-limited settings. Mamba-based architectures and contemporary large-kernel convolutional neural networks (CNN) effectively enlarge the receptive field; however, Mamba's state space model (SSM) relies on fixed sequential scanning orders (e.g., row-wise or column-wise rasterization) to serialize 2-D spatial data into 1-D sequences, which inherently imposes a rigid spatial traversal that cannot flexibly adapt to irregular spatial configurations or heterogeneous land-cover distributions prevalent in real-world hyperspectral scenes—causing important spatial dependencies between non-adjacent but semantically related pixels to be weakly captured or lost. Also, current ConvNet designs still don't perform well at modeling multiorder spectral–spatial interactions. To address these constraints, we introduce MogaHSI, an innovative series of convolutional networks explicitly designed for HSI classification, achieving an optimal balance between precision and computational efficiency. MogaHSI employs a multiorder gated aggregation (Moga) mechanism, along with lightweight yet expressive convolutional operations, to adaptively capture both local pixel-level details and larger spectral–spatial context. The framework uses adaptive channel-wise reallocation and multiorder spatial context aggregation to make feature representations across spectral dimensions more distinct and strong. Comprehensive experiments on four benchmark HSI datasets—HanChuan, HongHu, Houston, and Pavia University (PaviaU)—illustrate that MogaHSI consistently surpasses representative CNN-, Transformer-, and Mamba-based methodologies while necessitating fewer parameters. For example, MogaHSI raises the average accuracy (AA) from 95.86% to 98.10% and the overall accuracy (OA) from 95.74% to 97.83% on the PaviaU dataset. This sets a strong standard for quick and generalizable HSI classification.**

***Index Terms*—Channel aggregation block (CAB), hyperspectral imaging (HSI) classification, multiorder gated aggregation (Moga), spatial aggregation block (SAB).**

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The authors are with the Department of Computer Science, Khalifa University, Abu Dhabi, United Arab Emirates (e-mail: garasgaras@yahoo.com; hasan.almarzouqi@ku.ac.ae).

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I. Introduction

As EARTH observation (EO) technology advances, and an increasing volume of remote sensing (RS) data has become available. These data support a wide range of applications, including agricultural monitoring [1], urban planning [2], disaster management [3], and mineral finding [4]. Among the available RS modalities, hyperspectral imaging (HSI) has emerged as a particularly powerful tool, capturing hundreds of contiguous spectral bands across the visible-to-infrared range. This rich spectral resolution enables fine-grained material discrimination that conventional RGB sensors cannot achieve, making HSI classification a fundamental and widely studied task in remote sensing analysis. However, this spectral richness introduces significant challenges: high interband redundancy, irregular spatial distributions of land-cover classes, and the difficulty of modeling multiorder spectral–spatial dependencies simultaneously. Addressing these challenges within a single-source HSI framework—without relying on auxiliary modalities—remains an open problem, particularly under resource-constrained deployment conditions.

Deep learning has substantially advanced HSI classification, particularly in addressing the challenge of modeling complex spectral–spatial feature representations. Existing approaches broadly fall into two categories: traditional machine learning methods [5], [6] and deep learning-based techniques. Within the deep learning category, two dominant architectural families have emerged: models built on convolutional and graph neural network foundations [7], [8], [9], and those based on the Transformer paradigm [10], [11], [12]. convolutional neural network (CNN)-based methods demonstrate strong capability in capturing local spatial patterns through hierarchical convolution; however, their inherently limited receptive fields constrain their effectiveness in modeling dependencies across distant spatial locations. Transformer-based methods, by contrast, leverage self-attention to achieve global contextual representation, but this capability comes at the cost of quadratic computational complexity with respect to input sequence length, making them expensive for high-dimensional HSI data. Beyond architectural considerations, methodological advances from related remote sensing tasks have also informed spectral–spatial representation learning.

Once the transformer paradigm proved successful in natural language processing, it was progressively extended to visual problems. Unlike CNNs, transformers achieve stronger performance in HSI classification by modeling global spectral–spatial relationships and adaptively attending to the most informative regions of the input [13], [14], [15]. By utilizing self-attention mechanisms, transformers can capture

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long-range dependencies throughout the image [10]. However, this global modeling capability comes at a high cost: the computational complexity scales quadratically with input sequence length, imposing substantial memory and processing overhead that limits practical deployment, particularly for large-scale hyperspectral scenes with hundreds of spectral bands.

More recently, Mamba-based [16], [17] state-space models have emerged as a promising alternative, offering linear computational complexity while maintaining competitive long-range modeling capacity. However, as discussed earlier, their reliance on fixed sequential scanning orders makes them ill-suited for the irregular spatial structures and heterogeneous land-cover distributions characteristic of real-world HSI data. Similarly, large-kernel CNN expand the receptive field but still fall short in modeling multiorder spectral–spatial interactions that are critical for fine-grained pixel-level discrimination.

Despite the progress made across these architectural paradigms, several fundamental challenges in HSI classification remain unresolved as follows.

1) *Reconciliation of Global Modeling and Computational Efficiency:* Transformer models suffer from quadratic computational complexity that limits scalability, while CNN-based models are constrained by translational invariance and restricted receptive fields that hinder their ability to capture global contextual information and long-range spectral–spatial dependencies. Mamba-based models partially address complexity but introduce rigid spatial traversal that fails to adapt to irregular land-cover boundaries. A unified framework that balances expressiveness, spatial flexibility, and computational efficiency is therefore needed.
2) *Complexity in RS Imagery:* Hyperspectral scenes are inherently heterogeneous, characterized by irregular spatial patterns, highly variable spectral signatures, and diverse object geometries across different land-cover categories. Existing methods often assume regular spatial structures or uniform spectral contributions, limiting their generalization across diverse real-world scenes. Models must therefore exhibit strong adaptability to distributional and geometric variations in terrestrial observation objects while maintaining robust classification performance.

To address the challenges of HSI classification, including dense pixel-level prediction, spectral redundancy, and complex spatial structures, we propose MogaHSI. This lightweight convolutional framework incorporates multiorder convolutional aggregation as a modeling principle for HSI-oriented representation. MogaHSI adheres to an efficient CNN framework; however, its innovation lies in reconfiguring the explicit organization and use of spatial–spectral interactions for hyperspectral data, rather than merely implementing isolated architectural changes. MogaHSI uses a spatial aggregation block (SAB) to model complementary low-, mid-, and high-order spatial interactions. This enables it to adapt to irregular object shapes and multiscale land-cover patterns often observed in HSI scenes. This differs from other efficient CNN, such as MogaNet, which use a uniform expansion of the receptive field.

While MogaHSI draws inspiration from the multiorder gated aggregation (Moga) principle introduced in MogaNet, the architectural and methodological contributions of this work are substantially distinct and HSI-driven. First, MogaNet operates on patch-level token representations; MogaHSI replaces this entirely with a dense pixel-wise embedding pipeline that produces per-pixel spectral embeddings and predicts all labels in a single forward pass—a fundamental redesign motivated by the dense prediction requirements of HSI classification. Second, MogaNet [18] lacks a mechanism for spectral band redundancy, a defining challenge of hyperspectral data; we introduce the channel aggregation block (CAB), a new module that performs adaptive channel-wise reallocation via a learnable subtract-and-scale formulation, explicitly suppressing redundant spectral responses while amplifying discriminative ones. Third, all architectural decisions—normalization strategy, pooling removal, hidden dimension (HDim), and activation function—are motivated by and validated on HSI datasets rather than general vision benchmarks, revealing HSI-specific design constraints that do not arise in natural image classification. The contribution is therefore not an adaptation of MogaNet but a ground-up HSI classification framework that selectively adopts multiorder spatial aggregation as one component within a broader HSI-oriented design.

The article's contributions can be summarized as the following.

1) We propose a Moga module to capture interactions among features of different orders for classifying HSI. Moga effectively models complex spectral–spatial dependencies, enhances feature diversity, and improves the network's ability to discriminate between spectrally similar classes by adaptively combining information from multiple receptive fields via gated mechanisms.
2) We propose a SAB that accurately captures the spatial context and structural relationships among nearby pixels. This block improves the model's ability to preserve spatial continuity and boundary details by integrating information across different scales. This makes feature representation and overall classification accuracy better in complex hyperspectral scenes.
3) We present a CAB that dynamically integrates information from multiple feature channels. CAB improves feature discrimination and the network's ability to represent information while maintaining computational efficiency by learning how different channels depend on one another and focusing on the most informative responses.
4) We demonstrate that the developed network attains state-of-the-art (SOTA) performance with significantly fewer parameters through extensive experimentation on multiple benchmark HSI datasets. Additionally, an ablation study confirms the effectiveness of every module in the system.

We conducted analyses on four real-world HSI datasets. These results show that, in both performance and parameter

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count, our approach outperforms the top CNN-, Transformer-, and Mamba-based HSIs methods. In this study, heterogeneous spectral–spatial characteristics refer to the inherent diversity within HSI. This includes highly redundant yet unevenly informative spectral bands, as well as spatial structures with varying scales, shapes, and contextual dependencies. SAB addresses spatial heterogeneity and irregular object geometry, whereas CAB addresses spectral redundancy and uneven channel contributions. They all work together to form a single framework that addresses the problems discussed above.

II. Related Work

The associated research for this study will be divided into four main types: The CNN-based, the graph-based, the Transformer-based, and the Mamba-based methods.

A. CNN-Based Methods

First, Liu et al. [19] introduced a unique spatial–spectral enhancement and fusion network for HSI classification with limited labeled samples, designated SSEFN. The authors specifically devise a spatial–spectral enhancement approach (SSES) to augment feature recognition from both spatial and spectral viewpoints. Then, Zhang et al. [20] proposed a cascaded spatial crossattention network (CSCANet) for HSI classification. The authors develop a cascaded spatial crossattention module that first performs crossattention across local and global features within the spatial domain. Also, Sun et al. [21] developed an HSI classification approach utilizing the spatial–spectral-channel 3-D attention mechanism (3DAM-SSC) and a DenseNet enhanced with a channel attention mechanism (CAM), referred to as 3DAM-CAM DenseNet.

In addition, Feng et al. [22] proposed a hierarchical progressive fusion network (HPFN) based on Kolmogorov–Arnold networks. The suggested network comprehensively examines HSI information across three scales (pixel, patch, and global) while maintaining a lightweight structure. Moreover, Tong et al. [23] introduced a crossdomain few-shot learning system utilizing an adversarial dilated large-kernel attention (LKA) network. This model comprises an adversarial dilated LKA feature extractor that enhances the ability to capture sparse features. Additionally, Zhang et al. [24] proposed a global-local residual fusion network (GLRFNet) for HSI classification. The model employs multikernel feature projection to construct a feature pool prior to deep extraction, thereby enhancing information exchange among operators. Finally, Ma et al. [25] introduced a lightweight spatial–spectral deformable CNN (LS2DCNet) for UAV-based HSI classification. The network reduces computational cost and processing time while maintaining competitive classification performance.

CNN-based methods are good at extracting local features, but their small receptive fields make them unsuitable for modeling long-range dependencies. As a result, new models for capturing contextual information are needed.

B. Graph-Based Methods

To begin, Hanachi et al. [26] presented MV-GRL, which stands for multiview graph representation learning, as a method for HSI classification through the utilization of spectral–spatial graph neural networks. In addition, Jamali et al. [27] introduced an attention graph convolutional network (Attention-GCN) that addresses the previously listed deficiencies. The created model demonstrated a significant superiority over various CNN and vision transformer (ViT)-based models. Also, Wang et al. [28] proposed an enhanced hypergraph neural network (EHGNN) for HSI classification. The method introduces a key hypergraph to improve semantic representation and integrates CNN and HGNN architectures into a unified end-to-end framework, enhancing interpretability and modeling capability.

Moreover, Lu et al. [29] proposed SDGCP, a hybrid architecture that integrates separable deep graph convolution with a CNN and prototype learning for HSI classification, thereby enabling the extraction of rich global structural representations without adding trainable parameters. Building upon this, Cao et al. [30] introduced GLMBG, a broad graph convolutional network that incorporates both global and local manifold embedding strategies to enhance HSI classification performance. In GLMBG, the authors created two modules for feature extraction and classification. Additionally, Lei et al. [31] proposed a novel framework for HSI classification that integrates knowledge graphs (KGs) with symbolic reasoning and connectionist learning. In contrast to prior methodologies, this system combines a KG to capture explicit symbolic knowledge, thereby offering a formal definition of land-cover categories. Finally, Wang et al. [32] developed a spectral–spatial hypergraph convolutional network (S2HGCN) that concurrently represents long-range relationships and local spectral–spatial characteristics. HSI is initially processed by an attention feature enhancement module to mitigate interference noise arising from spectral redundancy and fluctuations.

When applied to large graphs, graph-based methods have drawbacks, including over-smoothing and high computational costs, despite their effectiveness at capturing structural dependencies. Thus, more scalable yet expressive approaches are needed.

C. Transformer-Based Methods

Initially, Wu et al. [33] developed a crosswindow spectral–spatial transformer model for HSI classification. The fused-MBConv module is employed to extract low-level spectral–spatial features from HSI. Second, the authors offer a cross-window attention module to model long-range spatial dependencies and acquire finer spectral–spatial properties. Also, Meng et al. [34] suggested a channel-reduced transformer (CR-Former) for HSI classification. To improve computational efficiency, the authors initially provide a cross-region tokenization (CRT) method. Then, Wei et al. [35] introduced a multiscale grouped feature fusion ViT (MGFF-

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ViT), engineered to augment spectral feature representation while preserving parameter efficiency.

Moreover, Tang et al. [36] proposed HyperEAST, an efficient dual-branch ViT framework that models spectral and spatial features separately. The method employs a linear fusion attention mechanism based on lightweight convolutions, replacing dot-product attention to achieve local–global representation learning with linear complexity. In addition, An et al. [37] developed a multiscale low-rank and sparse transformer (MLSFormer) that efficiently incorporates both low-rank and sparse principles. The authors use tensor low-rank decomposition (TLRD) to decompose the query, key, and value matrices into low-rank tensor products, thereby capturing predominant low-rank structures. Furthermore, Ding et al. [38] developed a lightweight CNN-Transformer framework based on specialized spectral–spatial blocks (SS-Blocks), which consist of spectral and spatial interaction modules (SpeIM and SpaIMs). Additionally, Wang et al. [39] introduced a multitask architecture that incorporates SR as a supplementary job to improve classification performance. The authors leverage the advantages of a CNN and a transformer to extract high- and low-frequency information, thereby constructing a dual-branch network and designing a frequency-domain interaction module (FIM) to enhance information exchange between the two branches, thereby improving representation capability.

Although transformer-based methods are reliable for modeling long-range relationships, their high computational cost makes them difficult to implement in resource-constrained environments.

D. Mamba-Based Methods

First, Zhuang et al. [40] introduced HSI classification using frequency-aware hierarchical Mamba (FAHM). FAHM consists of three primary components: an adaptive bidirectional Mamba block (ABMB), a frequency group embedding module (FGEM), and a spatial–spectral interaction block (SSIB). Also, Wang et al. [41] proposed MambaHSI+, which combines spectral trajectory learning with bidirectional state-space modeling. The architecture introduces recursive state transitions that enable multidirectional propagation of spatial information, further enhanced by reverse-order modeling. Then, Xu et al. [42] developed a method for improved HSI classification using sparse deformable Mamba (SDMamba), which has the following advantages. First, an effective sparse deformable sequencing (SDS) method is developed to adaptively learn an "optimal" sequence that improves the Mamba sequence.

In addition, Ding et al. [43] showed a multiscale spatial–spectral fusion network (M2S2F-Net) powered by Mamba. The spatial–spectral multigranularity feature extraction module of this network extracts spatial–spectral features at multiple granularities, and the spatial–spectral fusion attention module adaptively enhances spatial–spectral correlation. Furthermore, Paoletti et al. [44] addressed the parameter reduction issue by proposing an end-to-end DenseMixerMamba network for HSI classification that integrates Mamba, attention mechanisms, and dense architectures. Ultimately, Dou et al. [45] proposed that light detection and ranging (LiDAR)/synthetic aperture radar (SAR) data, as well as HSIs, can be jointly classified using a multilevel cross-modal attention-guided Mamba (MCAMamba) framework.

Mamba-based approaches explore large receptive fields while maintaining linear complexity, helping overcome the drawbacks of CNNs and Transformers. However, we managed to surpass the Mamba model by using a CNN layer of point and depth-wise convolution (DWConv)

III. Methodology

A. Motivation and Overall Idea

Since HSI classification is a pixel-by-pixel discriminative task, it presents a unique difficulty that requires precise modeling of subtle spectral–spatial distinctions in high-dimensional data. Long-range modeling of individual sequences can compromise the spatial integrity of the original image. Moreover, traditional CNN architectures are constrained by local receptive fields, limiting their ability to capture global contextual patterns. Although theoretically capable of simulating global interactions, transformer-based alternatives are hampered by the excessive quadratic processing complexity of self-attention methods $O(S^2)$, where S denotes the input sequence length. The granularity required for pixel-level distinction is lost in real implementations due to this complexity constraint, which forces coarse patch-level tokenization. The SAB and an adaptive CAB are the foundation of the encoder [18]. We will examine more closely the CAB, which can represent the reallocation of multiorder features, and the SAB, which can represent Moga.

B. Overview

Fig. 1 summarizes the proposed MogaHSI. The segmentation head, encoder backbone, and embedding layer are the framework's three primary parts.

1) Embedding Layer: The spectral vector is projected into an embedding space by the embedding layer. Unlike existing patch-based embedding techniques, our embedding layer produces an embedding for each pixel. The fine-grained pixel embeddings produced by this architecture are crucial for dense prediction applications such as HSI classification. In particular, HSI $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$ can be used to create the fine-grained pixel embedding $E \in \mathbb{R}^{H\times W\times D}$ in the manner described below

$$E = Embedding\,(I) = SiLU\,(GN\,(Conv\,(I))) \tag{1}$$

where Conv, group normalization (GN), and SiLU represent the convolutional layer with a 1×1 convolution, the SiLU activation function, and the GN layer, respectively. H, W, and C represent the height, width, and spectral channel number of the input HSI, respectively. D denotes the embedding dimension. $\mathbf{I}$ represents the input HSI and E is the resultant embedding.

2) Encoder Backbone: is utilized to supply distinct spatial–spectral information for classification. The encoder backbone comprises two primary components: a SAB and a channel aggregation layer (CAB) [18]. The procedure is broken down as follows:

$$\mathbf{F} = \text{Encoder}\,(E) \tag{2}$$

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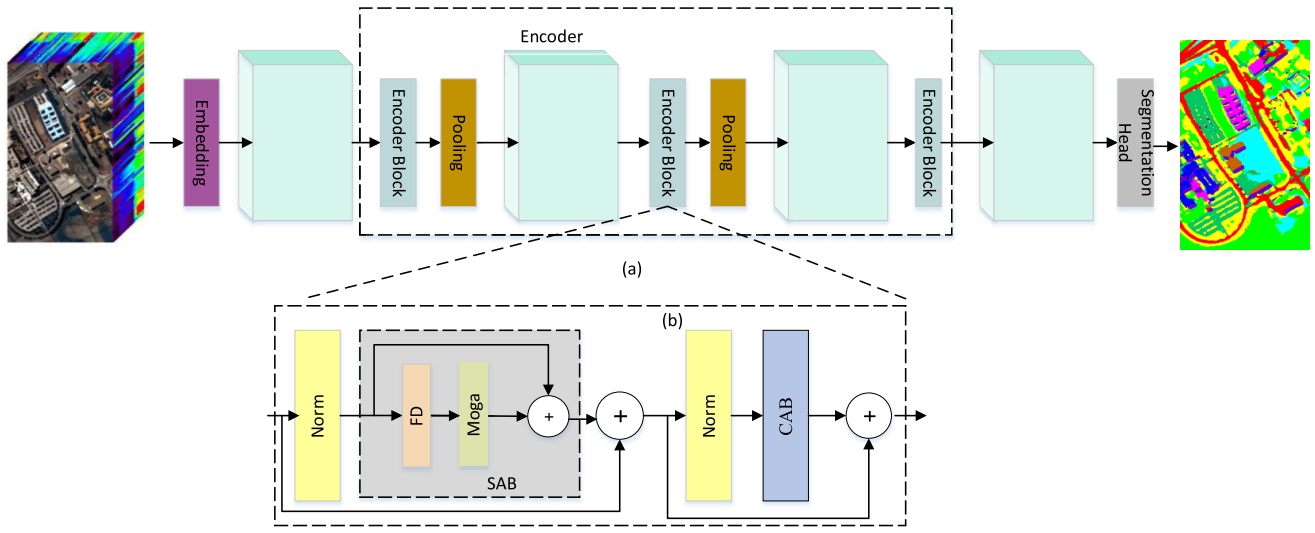


Fig. 1. (a) MogaHSI's general design. (b) Encoder block's structure. The SAB block contains the FD block and the moga block.

where **F** and the encoder represent the encoder backbone and the derived hidden features, respectively.

3) Segmentation Head: uses a convolutional layer with a 1×1 kernel size to get the final logits, which are expressed as $\mathbf{l} = \text{SegHead}(\mathbf{F})$.

C. Encoder Block

As illustrated in Fig. 1, the encoder block is designed using the Norm layer as well as the CAB and SAB. The complete Moga block function, assuming that the input feature is F_{i-1}, can be expressed as follows:

$$F_{i-1,s} = F_{i-1} + H_{SAB}\left(H_{Norm}\left(F_{i-1}\right)\right) \tag{3}$$

$$F_{i-1,c} = F_{i-1,s} + H_{CAB}\left(H_{Norm}\left(F_{i-1,s}\right)\right) \tag{4}$$

where the normalization layer's function is denoted by H_{Norm}.

1) Spatial Aggregation Block: The purpose of the SAB [18] is to combine multiorder contexts into a single design. This SAB block has two cascaded components, as seen in Fig. 2

$$F_{SAB} = X + H_{Moga}\left(H_{FD}\left(X\right)\right). \tag{5}$$

Let H_{Moga} denote the Moga module and H_{FD} represent the feature decomposition (FD) module [18]. The FD module [18] is built upon two complementary perception mechanisms, namely static and adaptive regionality, which together enable the extraction of multiorder features exclusively through convolutional operations. Rather than using all interaction orders, the module employs only 0- and n-order interactions, deliberately omitting m-order interactions. Specifically, n-order interactions, which capture intricate global shape information, are handled via Conv1 $\times$ 1$(\cdot)$, whereas the 0-order interaction, responsible for encoding common local texture patterns, is addressed through global average pooling (GAP$(\cdot)$). Furthermore, the FD$(\cdot)$ mechanism incorporates a dynamic filtering strategy that suppresses redundant or trivial interactions, thereby directing the model's focus toward the most informative multiorder interactions. Consequently, this module can be formulated as follows:

$$Y = H_{Conv1}\left(X\right) \tag{6}$$

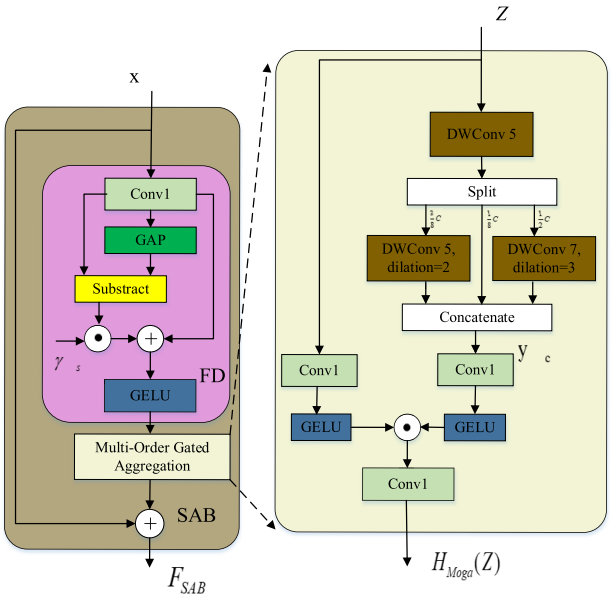


Fig. 2. SAB design.

$$Z = H_{GELU}\left(Y + \gamma_s \odot \left(Y - H_{GAP}\left(Y\right)\right)\right) \tag{7}$$

where $\gamma_s \in \mathbb{R}^{C\times 1}$ is a learnable scale factor initialized to zero, H_{Conv1} denotes the $\text{Conv}_{1\times1}$ operation, H_{GELU} denotes the GELU activation, and H_{GAP} denotes GAP. The function of the GELU activation is represented by H_{GELU}, and the GAP function is represented by H_{GAP}. Multiorder contexts are aggregated using the Moga module. There are three parallel DWConv layers used to capture low-, middle-, and high-order interactions. These layers have dilation ratios $d \in \{1, 2, 3\}$. Assume that the Moga's input feature is the FD's output feature. For low-order features, $Z \in \mathbb{R}^{C\times S}$, $DW_{5\times5,d=1}$ is employed first. The resulting output is then factorized along the channel dimension, where $C_l + C_m + C_h = C$, to produce $Z_l \in \mathbb{R}^{C_l\times S}$, $Z_m \in \mathbb{R}^{C_m\times S}$, and $Z_h \in \mathbb{R}^{C_h\times S}$ are then passed to

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$DW_{5\times5,d=2}$ and $DW_{7\times7,d=3}$, respectively, while Z_l simultaneously acts as an identity mapping. Finally, multiorder contexts $Y_C = \text{Concat}(Y_{l,C_l}, Y_m, Y_h)$ are formed by concatenating the outputs of Z_l, Z_m, and Z_h. It should be noted that, unlike those $DW_{7\times7}$ used in ConvNeXt [46]. The proposed FD(·) and multiorder DWConv layers achieve lower computational complexity.

Since the testing results demonstrated that GELU activation produced the highest performance, it is utilized in the gating branch to aggregate the output contexts from the context branch [47]. Consequently, the following is a representation of the Moga module's output:

$$F_{SAB} = H_{Conv1}\left(H_{GELU}\left(H_{Conv1}\left(X\right)\right) \odot H_{GELU}\left(H_{Conv1}\left(Y_c\right)\right)\right) + X. \tag{8}$$

The SAB's role can then be expressed as follows:

$$F_{SAB} = H_{SAB}\left(X\right) \tag{9}$$

where the SAB block's function is H_{SAB}.

Many more middle-order interactions can be captured by the MogaHSI when the SAB module is used. With its significantly more efficient parameters and FLOPs, this SAB block can produce good results without the need for expensive aggregation (e.g., self-attention).

Hyperspectral scenes show spatial patterns that are not uniform, such as object boundaries that are not straight and contextual dependencies that vary with scale. The SAB solves this problem by using Moga to combine low-, mid-, and high-order spatial contexts.

D. Channel Aggregation Block

The CAB is designed to directly address one of the main problems in learning to represent HSIs: the high redundancy among bands and the uneven contributions of different spectral channels. Hyperspectral data is made up of hundreds of highly correlated spectral bands, while natural RGB images are made up of only three colors. Only some of the channel interactions are useful for classification. Conventional convolutional or attention-based channel modeling methods often treat all channels equally or rely on computationally expensive attention mechanisms. This can obscure discriminative spectral cues and increase computational cost. Thus, CAB is introduced as a simple yet effective method for *adaptive channel-wise reallocation* based on the spectral properties of HSI data.

As shown in Fig. 3, CAB operates through a local-global channel-interaction pipeline. The block first applies a 3×3 depthwise convolution (DWConv), $H_{DW_{3\times3}}(\cdot)$, to capture local spatial context for each channel separately. This keeps spectral specificity while improving spatial awareness. After this operation, there is a pointwise projection $H_{Conv1}(\cdot)$ with a channel expansion ratio r. This makes the hidden feature space better able to represent complex structures and enables higher-order channel interactions. Using GELU to activate the intermediate feature representation gives

$$Y = H_{GELU}\left(H_{DW_{3\times3}}(H_{Conv1}\left(X\right))\right). \tag{10}$$

CAB adds a channel aggregation function, $H_{CA}(\cdot)$, to adaptively highlight useful spectral responses and filter out

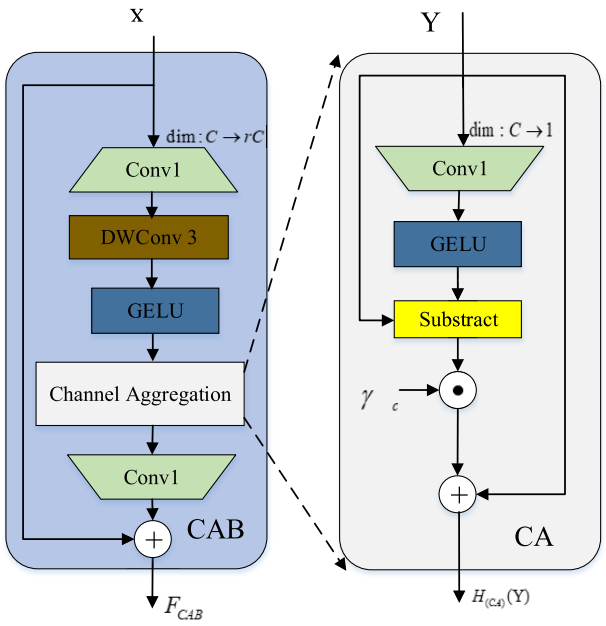


Fig. 3. CAB design.

unnecessary ones. Instead of directly learning channel attention weights, H_{CA} works by modeling the differences between channels in the expanded feature space. Specifically, a learnable channel-reduction projection $\mathbf{W}_r \in \mathbb{R}^{rC\times1}$, implemented as a 1×1 convolution followed by GELU activation, aggregates global channel responses across spatial dimensions and is applied as follows:

$$H_{CA}\left(Y\right) = Y + \gamma_c \odot \left(Y - H_{\text{GELU}}\left(\mathbf{W}_r \cdot Y\right)\right) \tag{11}$$

where $\gamma_c \in \mathbb{R}^{rC\times1}$ is a learnable per-channel scaling factor initialized to zero, consistent with γ_s in the FD module, and $\odot$ denotes element-wise multiplication. This formulation allows CAB to adjust the importance of channels by boosting those that provide complementary spectral information and deemphasizing those dominated by redundant or noisy responses.

Finally, a pointwise convolution is used to project the reweighted features back to the original channel dimension, and a residual connection is used to combine them with the input

$$F_{CAB} = H_{Conv1}(H_{CA}\left(Y\right)) + X. \tag{12}$$

This residual design stabilizes optimization and ensures that CAB selectively refines channel interactions without disrupting the original feature distribution.

Overall, CAB is essential to MogaHSI because it explicitly models spectral redundancy and variation in channel dependencies, which are specific to HSI. CAB improves feature discriminability while preserving the efficiency benefits of convolutional architectures by operating in a larger hidden space and performing adaptive channel-wise reallocation with minimal additional computational cost. In practice, we see that setting the channel expansion ratio to $r = 2$ strikes a good balance between performance gains and computational cost. This makes CAB especially useful for large-scale HSI classification tasks because spectral bands in HSI contribute differently; treating all channels equally may obscure important clues. The CAB models spectral heterogeneity directly by

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TABLE I

MODULE-LEVEL COMPLEXITY COMPARISON ON PAVIAU. † PATCH-LEVEL, ‡ IMAGE-LEVEL. SAB AND CAB ABLATION RESULTS ARE FROM TABLE X

Method	Complexity	Params (M)	FLOPs (G)	OA (%)
SpectralFormer†	$O(C^2S+CS^2)$	0.556	9086.19	91.10
GSC-ViT†	$O(C^2S+CS^2)$	0.364	2026.298	92.65
MambaHSI‡	$O(C^2S)$	5.777	39.18	95.74
MogaHSI w/o CAB‡	$O(C^2S)$	72.01	12.92	95.6
MogaHSI w/o SAB‡	$O(rC^2S)$	79.45	14.35	94.46
MogaHSI‡	$O(C^2S)$	134	24.15	97.83

changing the weights of channel responses in a larger feature space. This makes useful bands stronger and useless bands weaker.

E. Computational Complexity

We analyze the complexity of each proposed module with respect to the total number of spatial pixels $S = H_{\text{img}} \times W_{\text{img}}$ and channel dimension C.

1) SAB Complexity: The FD module applies a 1×1 convolution and GAP, costing $O(C^2S)$ and $O(CS)$, respectively. The Moga module applies three parallel DWConvs with kernel sizes $k \in \{5, 5, 7\}$ and dilation ratios $d \in \{1, 2, 3\}$, each costing $O(k^2CS)$. Since DWConvs operate channel-independently, the total Moga cost scales as $O(CS)$. Combined with two gating 1×1 convolutions at $O(C^2S)$, the overall SAB complexity is

$$\mathcal{O}_{\text{SAB}} = O\left(C^2S\right). \tag{13}$$

2) CAB Complexity: The CAB applies a 3×3 DWConv at $O(9CS)$, two 1×1 pointwise projections with an expansion ratio r at $O(rC^2S)$ each and a subtract-and-scale channel aggregation at $O(rCS)$. The total CAB complexity is

$$\mathcal{O}_{\text{CAB}} = O\left(rC^2S\right) \tag{14}$$

which with $r = 2$ remains linear in S.

3) Comparison With Self-Attention: Standard self-attention incurs $O(C^2S + CS^2)$ complexity, where the CS^2 term dominates for large HSI inputs. For example, at Pavia University (PaviaU) ($S = 610 \times 340 = 207\,400$), this renders image-level self-attention infeasible, as confirmed by our ablation, where even a reduced-dimension variant ($C = 8$) required significantly more memory and training time. In contrast, MogaHSI eliminates the quadratic term entirely

$$\mathcal{O}_{\text{MogaHSI}} = O\left(C^2S\right) \tag{15}$$

achieving spectral–spatial modeling at linear complexity. Table I summarizes the module-level comparison against representative attention-based methods.

F. Training and Inference

Algorithm 1 presents the complete training and inference procedure of MogaHSI. In contrast to patch-based HSI classification approaches [48], [49], which process local regions independently, MogaHSI performs dense prediction at the full-image level, generating class labels for every pixel in a single forward pass. The encoder leverages both the SAB and CAB modules to jointly capture spatial and spectral representations.

Algorithm 1 Training and Inference Pipeline of MogaHSI

// Training Stage

Input: Hyperspectral image $\mathbf{I}$ and corresponding label map $\mathbf{Y}_{tr}$

Initialization: Randomly set network parameters Θ

for $ep = 1, 2, \ldots, E$ **do**

// Forward pass over the entire image

$\mathbf{E} \leftarrow$ Embedding($\mathbf{I}$) // extract spectral–spatial embeddings

$\mathbf{F} \leftarrow$ Encoder($\mathbf{E}$) // includes SAB, CAB

$\mathbf{l} \leftarrow$ SegHead($\mathbf{F}$) // produce segmentation logits

// Loss calculation

$\mathcal{L} \leftarrow$ CrossEntropy($\mathbf{l}, \mathbf{Y}_{tr}$) // see Eq. (16)

// Parameter optimization

$\Theta \leftarrow$ Adam($\Theta, \nabla_\Theta \mathcal{L}$)

end for

Output: Optimized network parameters Θ **// Inference Stage**

Input: Test image $\mathbf{I}_{te}$

1) Load trained weights Θ.

2) Perform inference on $\mathbf{I}_{te}$ using the trained model: $\hat{\mathbf{y}} \leftarrow f_\Theta(\mathbf{I}_{te})$.

Output: Predicted label map $\hat{\mathbf{y}}$

The entire framework is optimized end-to-end in a single training stage. The training loss function $\mathcal{L}$ is given for the training as the following:

$$\mathcal{L} = \text{CrossEntropy}\left(\mathbf{l}, \mathbf{Y}_{tr}\right), \tag{16}$$

where $\mathbf{Y}_{tr}$ denotes the ground-truth label map and $\mathbf{l} = \text{MogaHSI}(\mathbf{I})$ represents the class logits produced by the network for the full input image $\mathbf{I}$. At inference time, given an unseen test image $\mathbf{I}_{test}$, the predicted label map $\hat{\mathbf{y}}$ is obtained by

$$\hat{y} = argmax\left(softmax\left(MogaHSI\left(I_{test}\right)\right)\right). \tag{17}$$

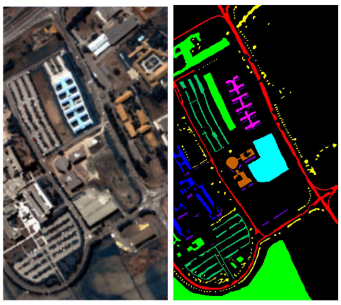
IV. EXPERIMENTS

This section outlines the experimental framework, covering the benchmark datasets, performance metrics, and implementation configurations adopted in this study. Subsequently, both quantitative and qualitative evaluations are conducted against leading existing methods. A systematic ablation study is also conducted to assess the individual contributions of each proposed component.

A. Experimental Setup

Datasets: To thoroughly evaluate the proposed method, we employ four widely recognized hyperspectral datasets, including PaviaU and Houston. WHU-Hi-HanChuan (HanChuan) [50] and WHU-Hi-HongHu (HongHu) [50].

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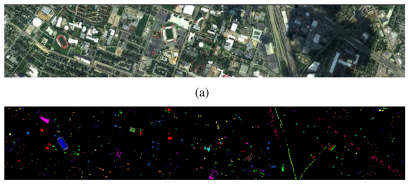


(a) (b)

ID	Color	Category	Train	validation	Test	Total
1		Asphalt	30	10	6591	6631
2		Meadows	30	10	18609	18649
3		Gravel	30	10	2059	2099
4		Trees	30	10	3024	3064
5		Metal sheets	30	10	1305	1345
6		Bare soil	30	10	4989	5029
7		Bitumen	30	10	1290	1330
8		Bricks	30	10	3642	3682
9		Shadows	30	10	907	947
Total			270	90	42416	42776

(c)

Fig. 4. Pavia dataset. (a) Image in false color. (b) Ground truth. (c) Sample and category settings.



(a)

(b)

ID	Color	Category	Train	validation	Test	Total
1		Healthy Grass	30	10	1211	1251
2		Stressed Grass	30	10	1214	1254
3		Synthetic Grass	30	10	657	697
4		Tree	30	10	1204	1244
5		Soil	30	10	1202	1242
6		Water	30	10	285	325
7		Residential	30	10	1228	1268
8		Commercial	30	10	1204	1244
9		Road	30	10	1212	1252
10		Highway	30	10	1187	1227
11		Railway	30	10	1195	1235
12		Parking Lot 1	30	10	1193	1233
13		Parking Lot 2	30	10	429	469
14		Tennis Court	30	10	388	428
15		Running Track	30	10	620	660
Total			450	150	14429	15029

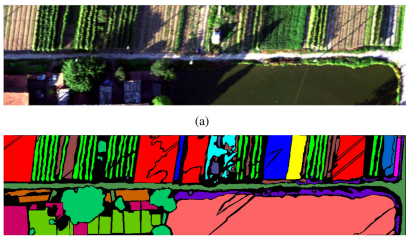
(c)

Fig. 5. Houston dataset. (a) Image in false color. (b) Ground truth. (c) Sample and category settings.

1) Pavia University: The PaviaU scene was collected by the ROSIS airborne sensor, producing a hyperspectral cube of 103 spectral bands over a spatial extent of 610 × 340 pixels. It contains 42 776 labeled samples distributed across nine distinct land-cover classes. The false-color image, corresponding ground-truth map, and sample distribution configuration are depicted in Fig. 4.

2) Houston: The Houston dataset captures the University of Houston campus and its surrounding area, recorded by the ITRES CASI 1500 HS sensor with 144 spectral bands at a spatial resolution of 349 × 1905 pixels. This scene was originally released as part of the 2013 Data Fusion Contest organized by the IEEE Geoscience and Remote Sensing Society (GRSS). It covers 16 land-cover categories spanning urban, commercial, and vegetation regions. Further details regarding the dataset are illustrated in Fig. 5.

3) HanChuan: The WHU-Hi-HanChuan scene was gathered on June 17, 2016, between 17:57 and 18:46, using a Headwall Nano-Hyperspec sensor with a 17 mm focal



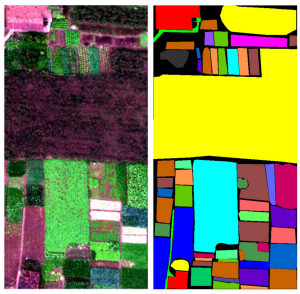
(a)

(b)

ID	Color	Category	Train	validation	Test	Total
1		Strawberry	30	10	44695	44735
2		Cowpea	30	10	22713	22753
3		Soybean	30	10	10247	10287
4		Sorghum	30	10	5313	5353
5		Water spinach	30	10	1160	1200
6		Watermelon	30	10	4493	4533
7		Greens	30	10	5863	5903
8		Trees	30	10	17938	17978
9		Grass	30	10	9429	9469
10		Red roof	30	10	10476	10516
11		Gray roof	30	10	16871	16911
12		Plastic	30	10	3639	3679
13		Bare soil	30	10	9076	9116
14		Road	30	10	18520	18560
15		Bright object	30	10	1096	1136
16		Water	30	10	75361	75401
Total			480	160	256890	257530

(c)

Fig. 6. HanChuan dataset. (a) Image in false color. (b) Ground truth. (c) Sample and category settings.



(a) (b)

ID	Color	Category	Train	validation	Test	Total
1		Red roof	30	10	14001	14041
2		Road	30	10	3472	3512
3		Bare soil	30	10	21781	21821
4		Cotton	30	10	163245	163285
5		Cotton firewood	30	10	6178	6218
6		Rape	30	10	44517	44557
7		Chinese cabbage	30	10	24063	24103
8		Pakchoi	30	10	4014	4054
9		Cabbage	30	10	10779	10819
10		Tuber mustard	30	10	12354	12394
11		Brassica parachinensis	30	10	10975	11015
12		Brassica chinensis	30	10	8914	8954
13		Small Brassica chinensis	30	10	22467	22507
14		Lactuca sativa	30	10	7316	7356
15		Celtuce	30	10	962	1002
16		Film covered lettuce	30	10	7222	7262
17		Romaine lettuce	30	10	2970	3010
18		Carrot	30	10	3177	3217
19		White radish	30	10	8672	8712
20		Garlic sprout	30	10	3446	3486
21		Broad bean	30	10	1288	1328
22		Tree	30	10	4000	4040
Total			660	220	385813	386693

(c)

Fig. 7. HongHu dataset. (a) Image in false color. (b) Ground truth. (c) Sample and category settings.

length mounted on a Leica Aibot X6 UAV V1 platform, over the Hanchuan region of Hubei Province, China. The scene encompasses a mixed rural-urban landscape featuring buildings, water bodies, and agricultural fields cultivating a variety of crops, including strawberry, cowpea, soybean, sorghum, water spinach, watermelon, and leafy vegetables. The acquired image spans 1217×303 pixels across 274 spectral bands within the 400–1000 nm wavelength range. A full description of this dataset is presented in Fig. 6.

4) HongHu: The WHU-Hi-HongHu scene was collected on November 20, 2017, between 16:23 and 17:37, by a Headwall Nano-Hyperspectral imaging sensor with a 17 mm focal length aboard a DJI Matrice 600 Pro UAV, over Honghu City, Hubei Province, China. The dataset is particularly distinctive in that it

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TABLE II

PAVIAU QUANTITATIVE ACCURACY (ACC% ± STD%). BEST IN BOLD AND SECOND IN UNDERLINE

Class	ML-based		GCN-based		CNN-based		Transformer-based		SSM-based	
	SVM TGRS2004	RF TGRS2005	DMSGer TNNLS2022	GiGCN TNNLS	FullyConNet TGRS2022	CLOLN TGRS2024	SpectralFormer TGRS2021	GSC-ViT TGRS2024	MambaHSI TGRS2024	MogaHSI Proposed
Asphalt	93.69 ± 2.34	93.74 ± 1.91	81.49 ± 4.78	90.36 ± 1.73	90.98 ± 7.20	**98.27 ± 1.00**	92.60 ± 2.29	97.10 ± 1.64	94.35 ± 1.73	95.29 ± 3.07
Meadows	88.56 ± 2.41	87.29 ± 1.93	87.60 ± 3.00	**98.86 ± 0.78**	89.87 ± 13.81	98.98 ± 0.76	93.58 ± 2.14	97.68 ± 1.32	96.43 ± 1.92	98.36 ± 1.63
Gravel	50.42 ± 5.13	48.47 ± 4.74	97.15 ± 1.28	91.66 ± 6.33	91.75 ± 5.61	87.83 ± 7.54	60.90 ± 8.15	86.46 ± 7.42	93.39 ± 5.35	**97.93 ± 3.09**
Trees	55.88 ± 5.67	55.21 ± 4.85	87.68 ± 2.85	81.34 ± 10.11	93.50 ± 6.74	**97.80 ± 1.46**	92.28 ± 6.23	91.83 ± 9.94	89.75 ± 3.60	95.33 ± 1.89
Metal sheets	95.67 ± 1.95	92.06 ± 1.78	99.38 ± 1.68	91.15 ± 13.11	99.54 ± 0.60	99.68 ± 0.59	98.00 ± 1.72	99.45 ± 0.53	**99.99 ± 0.02**	99.95 ± 0.14
Bare soil	49.11 ± 8.63	44.29 ± 5.43	99.66 ± 6.60	98.87 ± 1.90	92.36 ± 9.33	85.27 ± 7.50	63.92 ± 12.41	79.65 ± 9.24	98.60 ± 0.93	**99.36 ± 0.82**
Bitumen	41.46 ± 3.09	48.42 ± 4.67	99.36 ± 0.88	99.28 ± 0.61	96.34 ± 6.58	85.92 ± 7.00	48.11 ± 8.47	86.63 ± 10.10	96.50 ± 3.13	**99.81 ± 0.27**
Bricks	73.17 ± 4.61	70.94 ± 6.73	88.65 ± 2.94	87.68 ± 7.46	89.67 ± 4.82	90.93 ± 4.12	81.22 ± 4.66	89.50 ± 4.70	94.40 ± 2.93	**97.91 ± 1.03**
Shadows	**99.91 ± 0.08**	99.81 ± 0.11	96.89 ± 0.82	97.21 ± 3.36	96.73 ± 9.02	97.12 ± 2.03	88.78 ± 5.92	98.71 ± 1.10	99.36 ± 0.62	98.93 ± 0.81
OA(%)	71.31 ± 3.15	70.68 ± 2.61	89.55 ± 1.22	94.47 ± 0.97	91.31 ± 9.36	94.97 ± 1.43	91.10 ± 2.60	92.65 ± 2.30	95.74 ± 0.90	**97.83 ± 0.68**
AA(%)	71.98 ± 1.39	71.14 ± 1.61	93.15 ± 0.70	92.93 ± 1.07	93.41 ± 6.57	93.53 ± 1.59	79.93 ± 2.67	91.89 ± 1.73	95.86 ± 1.11	**98.1 ± 0.58**
Kappa(%)	63.96 ± 3.19	63.13 ± 2.81	86.49 ± 1.54	92.67 ± 1.26	93.12 ± 5.19	93.38 ± 1.84	77.40 ± 4.41	90.38 ± 2.95	95.00 ± 2.24	**98.63 ± 1.31**

TABLE III

HOUSTON DATASET QUANTITATIVE ACCURACY (ACC% ± STD%)

Class	ML-based		GCN-based		CNN-based		Transformer-based		SSM-based	
	SVM TGRS2004	RF TGRS2005	DMSGer TNNLS2022	GiGCN TNNLS	FullyConNet TGRS2022	CLOLN TGRS2024	SpectralFormer TGRS2021	GSC-ViT TGRS2024	MambaHSI TGRS2024	MogaHSI Proposed
Healthy Grass	93.68 ± 5.88	93.00 ± 0.45	91.27 ± 4.59	83.21 ± 5.03	92.69 ± 7.57	88.24 ± 0.90	91.29 ± 5.38	90.16 ± 6.48	**95.20 ± 4.66**	92.2 ± 4.03
Stressed Grass	85.92 ± 6.74	91.95 ± 4.53	92.62 ± 1.87	91.23 ± 3.49	94.61 ± 2.42	97.41 ± 2.77	94.61 ± 2.54	97.41 ± 1.26	**98.21 ± 1.26**	96.22 ± 1.91
Synthetic Grass	98.83 ± 1.38	97.82 ± 3.65	99.23 ± 0.96	98.83 ± 1.66	99.21 ± 0.74	95.21 ± 9.80	94.68 ± 3.25	**99.83 ± 0.62**	99.74 ± 0.46	99.89 ± 0.18
Tree	**98.89 ± 0.50**	97.42 ± 1.20	97.84 ± 1.69	96.83 ± 0.65	98.14 ± 1.50	97.42 ± 0.65	98.13 ± 1.62	97.62 ± 1.49	98.54 ± 0.56	97.47 ± 2.
Soil	90.87 ± 2.41	87.75 ± 4.04	98.67 ± 1.79	93.14 ± 4.62	97.07 ± 6.99	97.07 ± 2.66	98.18 ± 7.22	97.63 ± 2.69	99.54 ± 0.76	**99.68 ± 0.47**
Water	87.86 ± 4.04	86.03 ± 2.64	**99.77 ± 0.63**	99.77 ± 0.63	97.48 ± 2.84	97.48 ± 3.06	98.63 ± 1.69	99.54 ± 0.62	99.31 ± 0.93	98.88 ± 1.62
Residential	73.99 ± 4.46	74.97 ± 4.97	83.13 ± 2.53	82.34 ± 5.24	86.22 ± 4.41	94.02 ± 2.94	91.47 ± 3.54	93.11 ± 2.39	92.35 ± 1.98	**94.26 ± 3.33**
Commercial	79.40 ± 11.34	83.92 ± 3.22	67.40 ± 3.09	66.85 ± 5.84	83.01 ± 9.92	83.01 ± 3.08	84.41 ± 7.91	90.53 ± 3.41	**93.51 ± 2.98**	86.4 ± 5.59
Road	82.71 ± 4.68	85.76 ± 2.27	94.88 ± 1.02	94.04 ± 1.99	94.68 ± 2.49	95.21 ± 0.69	95.22 ± 2.42	96.53 ± 0.92	**96.83 ± 1.08**	93.23 ± 4.55
Highway	70.50 ± 4.76	73.69 ± 5.63	98.16 ± 1.84	92.31 ± 2.69	93.38 ± 6.32	94.76 ± 1.54	93.38 ± 2.48	95.22 ± 3.20	95.55 ± 1.42	**99.76 ± 0.38**
Railway	74.60 ± 5.43	73.47 ± 6.43	95.82 ± 0.95	89.20 ± 2.56	91.35 ± 8.94	93.53 ± 2.97	93.53 ± 3.54	95.83 ± 1.97	97.45 ± 0.92	96.6 ± 2.44
Parking Lot 1	64.23 ± 4.99	58.37 ± 5.03	81.28 ± 1.89	90.45 ± 3.63	88.89 ± 4.64	91.11 ± 3.15	90.00 ± 5.19	91.85 ± 1.54	**90.18 ± 1.72**	89.97 ± 4.59
Parking Lot 2	46.43 ± 10.62	31.31 ± 8.95	96.95 ± 2.07	98.30 ± 2.67	100.00 ± 0.00	91.11 ± 3.15	95.56 ± 6.67	100.00 ± 0.00	100.00 ± 0.00	97.32 ± 2.59
Tennis Court	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	98.30 ± 2.67	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.9 ± 0.31
Running Track	99.16 ± 0.46	97.81 ± 1.84	99.76 ± 0.19	99.76 ± 0.19	98.47 ± 3.70	98.37 ± 3.06	98.62 ± 1.12	99.77 ± 0.63	**100.00 ± 0.00**	96.03 ± 2.82
OA(%)	79.63 ± 0.48	78.98 ± 0.37	90.36 ± 1.29	88.91 ± 1.26	91.38 ± 2.36	91.48 ± 1.56	91.82 ± 0.71	93.77 ± 0.58	94.46 ± 0.37	**95.19 ± 0.7**
AA(%)	80.70 ± 1.03	79.24 ± 0.83	92.17 ± 1.07	89.85 ± 1.91	91.93 ± 3.55	91.80 ± 1.46	89.90 ± 1.52	92.97 ± 0.74	95.34 ± 0.78	**95.85 ± 0.72**
Kappa(%)	77.97 ± 1.12	77.25 ± 0.98	89.58 ± 1.39	87.16 ± 2.44	90.64 ± 2.68	89.52 ± 1.87	88.22 ± 0.79	91.81 ± 1.14	94.21 ± 1.9	**95.5 ± 1.6**

encompasses a diverse range of crop types, including multiple cultivars belonging to the same species. The resulting image covers a spatial extent of 940 × 475 pixels and contains 270 spectral bands spanning the 400–1000 nm wavelength range. Further details regarding this dataset are provided in Fig. 7.

a) Evaluation Metrics: Three evaluation measures commonly employed in hyperspectral classification were utilized in the experiments: overall accuracy (OA), average accuracy (AA), and kappa coefficient (k). To reduce the degree of unpredictability introduced by the random selection of training samples, the primary comparative experiments and ablation investigations were conducted ten times. An increased mean value and a reduced standard deviation of these variables indicate improved model performance.

B. Comparison Methods

We evaluate multiple approaches for HSI classification.

1) Regarding ML-Based:

a) SVM[TGRS2004] [5]: It employs support vector machines for the classification of HSI.

b) RF[TGRS2005] [6]: The model classifies HSI data utilizing RF.

2) For GCN Based:

a) DMSGER[tnnls2022] [51]: The model employs a dynamic multiscale graph convolutional classifier to simultaneously extract discriminative features at both the pixel and region levels.

b) GIGCN[tnnls2022] [52]: Classification performance is enhanced by jointly leveraging information derived from both internal and external superpixel representations.

3) For CNN Based:

a) Fullycontnet[tgrs22] [53]: The framework integrates a multiscale feature extraction strategy based on pyramid pooling with a scale-aware attention mechanism to generate more discriminative classification representations, taking the raw HSI data directly as input.

b) CLOLN[TGRS24] [49]: The model is a streamlined network aimed at channel layers, intended to reduce computational complexity and the number of parameters in CNNs.

4) For Transformer Based:

a) Spectralformer [TGRS21] [11]: Spectrally localized sequential information is extracted from neighboring spectral bands to construct group-wise spectral embeddings tailored to the unique characteristics of hyperspectral data.

b) GSC-VIT [TGRS24] [48]: Local and global spectral–spatial representations are jointly captured through a groupwise separable convolution-based ViT architecture designed for HSI classification.

5) For Mamba-Based:

a) Mambahsi [TGRS24] [54]: The model utilizes Spatial–Spectral Mamba, incorporating the spatial Mamba block (SpaMB) and the spectral Mamba block (SpeMB).

6) Implementation Details: The proposed MogaHSI framework is implemented using the PyTorch deep learning library. To account for variability introduced by random sample selection and parameter initialization, all experiments are independently repeated ten times. In each trial, 30 and 10 samples per class are randomly selected to form the training and validation sets, respectively, with all remaining labeled samples reserved for testing. Since both MogaHSI and

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TABLE IV

HANCHUAN DATASET QUANTITATIVE ACCURACY (ACC% ± STD%) O

Class	ML-based		GCN-based		CNN-based		Transformer-based		SSM-based	
	SVM TGRS2004	RF TGRS2005	DMSGer TNNLS2022	GiGCN TNNLS	FullyConNet TGRS2022	CLOLN TGRS2024	SpectralFormer TGRS2021	GSC-ViT TGRS2024	MambaHSI TGRS2024	MogaHSI Proposed
Strawberry	92.03 ± 1.24	87.87 ± 1.73	74.51 ± 5.53	89.34 ± 2.23	78.59 ± 12.03	**95.97 ± 1.87**	95.13 ± 1.25	95.26 ± 1.50	89.72 ± 5.75	88.21 ± 4.83
Cowpea	69.50 ± 7.79	57.02 ± 4.42	62.51 ± 8.18	81.64 ± 3.38	84.21 ± 4.34	**92.48 ± 2.56**	87.27 ± 3.37	91.22 ± 3.05	81.37 ± 5.69	87.09 ± 5.89
Soybean	47.41 ± 4.94	45.31 ± 2.24	94.63 ± 2.80	81.06 ± 6.56	79.85 ± 9.55	80.03 ± 9.18	70.36 ± 6.15	78.37 ± 7.47	**94.88 ± 3.73**	91.02 ± 3.26
Sorghum	74.49 ± 9.57	51.50 ± 2.59	99.51 ± 0.50	67.09 ± 4.82	84.68 ± 13.06	91.12 ± 7.72	88.62 ± 6.67	85.17 ± 12.38	98.46 ± 0.54	**99.57 ± 0.61**
Water spinach	9.72 ± 0.77	9.25 ± 0.78	99.18 ± 0.92	34.98 ± 2.76	68.38 ± 27.44	56.82 ± 12.89	42.01 ± 11.48	46.57 ± 13.03	99.54 ± 0.85	**99.9 ± 0.16**
Watermelon	15.28 ± 2.00	11.66 ± 1.03	**94.18 ± 2.43**	53.81 ± 9.81	9.39 ± 15.90	38.07 ± 8.70	34.68 ± 6.40	35.97 ± 7.01	76.99 ± 8.7	82.82 ± 5.58
Greens	36.69 ± 2.30	45.58 ± 3.46	**98.72 ± 1.26**	80.84 ± 8.87	92.94 ± 4.47	68.87 ± 17.46	67.69 ± 10.29	66.90 ± 15.67	92.61 ± 3.34	95.43 ± 3.89
Trees	65.71 ± 5.51	66.22 ± 4.84	84.57 ± 4.09	89.27 ± 4.80	**91.24 ± 1.96**	81.52 ± 7.87	83.17 ± 4.16	82.39 ± 5.23	70.73 ± 7.10	85.14 ± 4.1
Grass	47.93 ± 7.71	40.87 ± 7.40	72.25 ± 5.64	65.14 ± 6.79	70.48 ± 8.43	68.59 ± 6.61	63.00 ± 4.53	60.64 ± 17.62	89.01 ± 3.62	**93.25 ± 3.14**
Red roof	74.50 ± 11.85	59.59 ± 6.64	87.81 ± 3.96	93.03 ± 4.21	85.23 ± 6.83	90.79 ± 14.71	88.76 ± 12.99	93.22 ± 4.31	94.44 ± 2.18	**96.49 ± 4.88**
Gray roof	35.90 ± 1.90	54.20 ± 9.34	**96.13 ± 1.66**	93.86 ± 5.06	92.58 ± 4.29	87.86 ± 4.63	77.41 ± 13.19	82.13 ± 7.96	89.56 ± 4.91	94.37 ± 4.36
Plastic	13.49 ± 2.57	17.50 ± 2.66	**99.09 ± 1.04**	89.22 ± 7.45	39.55 ± 27.62	63.48 ± 14.80	51.40 ± 6.89	51.92 ± 6.01	88.83 ± 6.91	92.92 ± 4.69
Bare soil	27.36 ± 3.66	33.89 ± 4.49	82.67 ± 3.9	66.66 ± 6.60	82.37 ± 2.46	50.21 ± 6.71	47.62 ± 9.17	53.28 ± 10.88	78.69 ± 3.73	**84.87 ± 6.03**
Road	80.55 ± 8.56	76.20 ± 6.53	75.85 ± 3.16	91.25 ± 2.43	88.31 ± 4.97	90.78 ± 2.74	88.92 ± 3.08	90.64 ± 3.07	**99.68 ± 3.00**	93.74 ± 2.73
Bright object	33.15 ± 10.97	26.80 ± 10.82	**98.36 ± 2.21**	46.57 ± 17.42	97.30 ± 2.18	62.42 ± 10.67	52.51 ± 8.59	70.05 ± 12.35	94.42 ± 2.41	97.12 ± 1.72
Water	**99.95 ± 0.06**	98.82 ± 1.33	92.23 ± 5.31	99.75 ± 0.14	93.64 ± 4.75	98.97 ± 1.02	99.47 ± 0.33	99.77 ± 0.23	98.28 ± 1.62	98.62 ± 1.24
OA(%)	60.98 ± 1.35	64.09 ± 1.78	84.39 ± 1.83	86.78 ± 1.43	84.88 ± 3.02	85.14 ± 2.93	82.99 ± 2.31	83.14 ± 3.96	90.21 ± 1.67	**92.73 ± 1.2**
AA(%)	51.48 ± 1.46	48.89 ± 1.15	88.26 ± 0.89	76.47 ± 1.62	77.42 ± 3.61	76.13 ± 3.26	71.13 ± 2.56	73.97 ± 2.06	89.20 ± 1.36	**92.53 ± 0.63**
Kappa(%)	56.00 ± 1.49	59.05 ± 1.86	82.01 ± 2.02	84.64 ± 1.62	75.80 ± 4.70	82.75 ± 3.31	80.29 ± 2.61	80.57 ± 4.33	89.07 ± 2.31	**91.73 ± 2.59**

TABLE V

HONGHU DATASET QUANTITATIVE ACCURACY (ACC% ± STD%)

Class	ML-based		GCN-based		CNN-based		Transformer-based		SSM-based	
	SVM TGRS2004	RF TGRS2005	DMSGer TNNLS2022	GiGCN TNNLS	FullyConNet TGRS2022	CLOLN TGRS2024	SpectralFormer TGRS2021	GSC-ViT TGRS2024	MambaHSI TGRS2024	MogaHSI Proposed
Red roof	94.30 ± 1.90	84.35 ± 5.74	92.96 ± 4.02	90.93 ± 2.64	96.78 ± 1.57	95.36 ± 5.56	96.66 ± 1.63	**98.16 ± 1.08**	96.47 ± 1.50	96.15 ± 1.2
Road	53.77 ± 4.54	52.01 ± 5.17	93.34 ± 4.05	51.88 ± 7.77	95.63 ± 2.54	78.60 ± 6.19	62.84 ± 4.24	73.78 ± 9.09	95.43 ± 2.33	**97.54 ± 1.05**
Bare soil	90.70 ± 3.65	90.92 ± 2.70	92.15 ± 3.46	78.80 ± 2.87	92.23 ± 1.51	**98.22 ± 1.32**	95.60 ± 2.81	97.21 ± 1.81	94.66 ± 1.05	93.5 ± 1.49
Cotton	97.42 ± 0.34	97.09 ± 0.51	60.26 ± 10.53	99.37 ± 0.32	93.59 ± 2.92	**99.48 ± 0.37**	99.42 ± 0.30	99.57 ± 0.14	96.96 ± 1.79	98.93 ± 0.78
Cotton firewood	7.61 ± 2.78	14.17 ± 2.64	99.30 ± 0.79	62.41 ± 8.43	96.03 ± 2.60	41.33 ± 18.92	42.14 ± 10.33	37.03 ± 9.33	96.20 ± 0.54	**97.13 ± 1.56**
Rape	90.59 ± 1.93	84.04 ± 1.60	84.36 ± 4.27	89.61 ± 3.17	91.61 ± 2.28	**96.09 ± 1.67**	95.73 ± 1.02	95.72 ± 1.83	93.90 ± 2.47	95.21 ± 2.12
Chinese cabbage	78.60 ± 3.21	72.03 ± 4.93	69.44 ± 3.44	82.91 ± 3.07	82.52 ± 3.43	**92.72 ± 1.96**	90.05 ± 2.26	90.38 ± 3.66	86.60 ± 3.30	86.95 ± 3.84
Pakchoi	13.56 ± 2.31	9.13 ± 2.07	**98.36 ± 2.87**	75.44 ± 8.66	96.81 ± 2.03	46.44 ± 6.02	35.59 ± 5.04	32.12 ± 4.83	96.08 ± 2.90	96.73 ± 3.56
Cabbage	96.57 ± 1.71	90.60 ± 3.89	95.14 ± 1.73	85.39 ± 3.04	96.62 ± 1.20	**97.04 ± 2.05**	97.89 ± 1.07	97.06 ± 2.73	94.78 ± 1.69	97.44 ± 1.6
Thick mustard	50.25 ± 7.08	33.29 ± 6.19	87.22 ± 4.16	92.42 ± 2.80	90.08 ± 2.81	84.88 ± 3.47	78.31 ± 4.58	79.23 ± 7.53	90.43 ± 2.72	**95.36 ± 1.4**
Brassica parachinensis	31.70 ± 5.48	25.71 ± 4.48	82.08 ± 7.09	75.17 ± 4.50	86.27 ± 3.66	72.03 ± 8.35	69.29 ± 7.09	72.41 ± 8.95	91.96 ± 2.87	**93.59 ± 3.38**
Brassica chinensis	42.58 ± 4.37	40.77 ± 3.36	**91.91 ± 2.46**	82.52 ± 5.36	88.62 ± 4.09	66.63 ± 8.82	58.68 ± 5.96	65.61 ± 8.66	87.40 ± 5.22	92.65 ± 3.48
Small Brassica chinensis	59.05 ± 4.52	54.24 ± 3.33	73.10 ± 6.40	**89.28 ± 2.70**	89.16 ± 2.24	82.67 ± 5.83	76.20 ± 6.07	77.49 ± 8.47	89.12 ± 3.12	88.56 ± 2.63
Lactuca sativa	67.70 ± 6.77	57.66 ± 7.93	94.87 ± 2.43	68.03 ± 5.44	93.88 ± 1.93	87.59 ± 7.03	83.89 ± 6.15	86.26 ± 11.93	94.01 ± 1.65	**95.75 ± 1.72**
Celery	8.36 ± 4.25	3.78 ± 1.16	**99.85 ± 0.28**	21.65 ± 2.65	98.42 ± 1.16	75.88 ± 12.93	61.24 ± 13.17	63.62 ± 20.25	98.36 ± 1.17	99.37 ± 0.61
Film covered lettuce	87.85 ± 2.55	80.73 ± 7.57	96.89 ± 1.26	71.70 ± 6.79	88.96 ± 4.28	97.39 ± 1.73	95.23 ± 2.99	97.42 ± 1.66	96.83 ± 1.60	**98.26 ± 1.01**
Romaine lettuce	55.60 ± 4.87	54.14 ± 6.89	99.23 ± 1.01	54.94 ± 4.04	95.10 ± 4.52	80.09 ± 9.70	68.68 ± 8.79	84.92 ± 6.09	98.60 ± 1.49	**99.61 ± 0.66**
Carrot	28.46 ± 3.59	18.32 ± 1.20	**98.44 ± 2.49**	67.65 ± 8.17	97.22 ± 1.43	65.55 ± 15.89	60.99 ± 7.01	67.44 ± 10.50	96.72 ± 1.82	98.06 ± 1.23
White radish	65.15 ± 5.12	47.85 ± 5.57	88.34 ± 4.02	72.71 ± 8.86	89.35 ± 3.28	82.53 ± 7.23	82.77 ± 4.78	85.10 ± 8.11	91.30 ± 3.07	**93.75 ± 1.83**
Garlic sprout	37.85 ± 5.53	17.29 ± 4.70	**99.02 ± 0.85**	66.48 ± 6.47	98.34 ± 1.10	81.38 ± 6.60	71.18 ± 8.78	71.64 ± 10.46	99.18 ± 0.42	98.51 ± 0.87
Broad bean	9.35 ± 0.92	9.68 ± 0.83	**99.96 ± 0.12**	65.20 ± 14.58	99.95 ± 0.12	30.19 ± 12.06	28.91 ± 8.18	25.92 ± 7.91	99.91 ± 0.18	99.58 ± 1.26
Tree	21.53 ± 1.27	20.00 ± 1.27	99.88 ± 0.24	85.35 ± 8.84	99.11 ± 1.00	55.02 ± 10.89	58.01 ± 15.28	46.26 ± 11.15	99.54 ± 0.51	**99.76 ± 0.3**
OA(%)	68.77 ± 1.27	59.85 ± 2.13	75.43 ± 4.34	87.22 ± 0.60	92.19 ± 1.35	87.93 ± 2.11	86.04 ± 2.83	86.18 ± 1.86	94.58 ± 1.01	**96.04 ± 0.52**
AA(%)	54.50 ± 0.41	48.08 ± 1.05	90.73 ± 0.92	74.08 ± 1.50	93.47 ± 0.48	77.60 ± 2.25	73.17 ± 2.45	74.74 ± 1.64	94.00 ± 0.61	**96.02 ± 0.38**
Kappa(%)	62.71 ± 1.28	53.22 ± 1.95	71.59 ± 4.32	83.95 ± 0.73	93.86 ± 1.54	85.06 ± 2.49	82.72 ± 3.22	82.93 ± 2.16	94.00 ± 1.88	**94.9 ± 1.0**

MambaHSI operate on the full image and predict all pixel labels in a single forward pass, the batch size is fixed at 1. The network is optimized using the Adam optimizer with a learning rate of 3×10^{-4}. The HDim D is set to 64, the number of encoder blocks is set to 3, and the CAB expansion ratio is set to 2. GN and the GELU activation function are adopted throughout the network, while `att_force_fp32` is disabled. All experiments are conducted on the same hardware platform, consisting of an NVIDIA Quadro RTX 8000 GPU, a 64-core Intel Xeon Gold 6258R CPU, and 92 GB of RAM.

C. Comparison With SOTAs

To comprehensively evaluate the effectiveness of the proposed method, we compared it with several SOTA HSI classification algorithms from both quantitative and qualitative perspectives. The selected baselines include several model architectures, such as ML-, GCN-, CNN-, and Transformer-based models, as well as Mamba alternative approaches. The corresponding quantitative and qualitative results are presented in Tables II–V and Figs. 8–11, respectively.

1) Pavia University: A comprehensive quantitative evaluation of competing models on the PaviaU dataset is shown in Table II, where the proposed MogaHSI demonstrates consistently superior performance. Across all three evaluation metrics, namely, OA, AA, and the Kappa coefficient, MogaHSI achieves the highest scores among all compared approaches. It achieves the highest OA of 97.83 ± 0.68, outperforming the third-best method (MambaHSI) by 2.09 percentage points, demonstrating clear superiority in classification performance. Furthermore, for AA, our method achieves 98.10 ± 0.58, outperforming the second-best method (GSC-VIT) by 6.21 percentage points, highlighting its robustness and reliability across diverse categories.

2) Houston Dataset: As evidenced by the results in Table III, MogaHSI attains leading performance across all evaluation criteria on the Houston benchmark. Specifically, the proposed method achieves an OA of 95.19% ± 0.70%, surpassing the second-ranked competitor, MambaHSI (94.46% ± 0.37%) by a notable margin, reflecting stronger discriminative capacity across spectrally diverse land-cover

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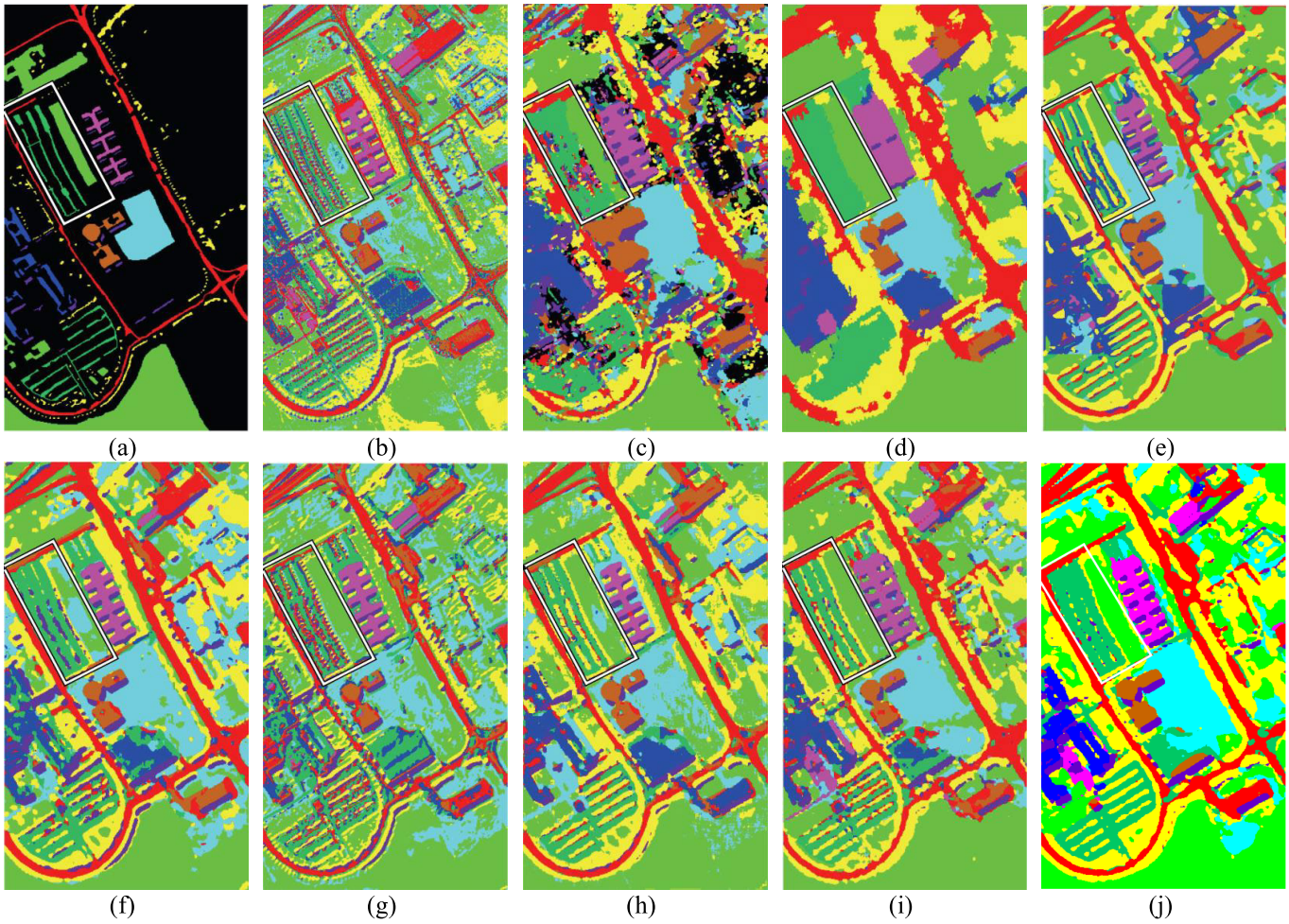


Fig. 8. Classification map of the Pavia dataset is presented qualitatively. (a) Map of verified accuracy. (b) Support vector machine. (c) DMSGer. (d) GiGCN. (e) FullyContNet. (f) CLOLN. (g) Spectralformer. (h) GSC-ViT. (i) MambaHSI. (j) MogaHSI. Fig. 4(c) serves as the basis for color interpretation.

categories. A similar trend is observed in terms of the Kappa coefficient, where MogaHSI records 95.5 ± 1.6, outperforming MambaHSI's 94.21 ± 1.9.

3) HanChuan Dataset: The results reported in Table IV demonstrate that MogaHSI establishes a new SOTA performance on the HanChuan dataset across all evaluation criteria, attaining an OA of $92.73\% \pm 1.2\%$, an AA of $90.45\% \pm 0.96\%$, and a Kappa coefficient of 92.53 ± 0.63. Relative to MambaHSI, the proposed method yields consistent gains of 2.51, 3.33, and 2.64 percentage points in OA, AA, and Kappa, respectively, underscoring its enhanced per-class discrimination and overall classification reliability. These gains indicate enhanced per-class discrimination and improved classification reliability.

4) HongHu Dataset: The proposed technique exhibits remarkable performance on the HongHu dataset across three principal evaluation metrics, achieving SOTA performance with significant margins. As shown in Table V, our model achieves an OA of 96.04 ± 0.52, exceeding the second-best competitor by 1.46 percentage points. The method achieves a Kappa coefficient of 94.9 ± 1.0, surpassing existing methods by 0.9 percentage points. These substantial improvements not only validate the method's superior discriminative performance across diverse categories but also highlight its exceptional stability in cross-validation settings, establishing new benchmarks.

The qualitative classification maps presented in Figs. 8–11 reveal that MogaHSI consistently yields more spatially coherent predictions with sharper object boundaries and notably fewer misclassified pixels compared to all competing methods. For example, as shown in Fig. 11, methods such as SVM, SpectralFormer, GSC-ViT, and CLOLN exhibit considerable salt-and-pepper noise across the HongHu scene, whereas DMSGer, GiGCN, and FullyContNet tend to produce poorly defined class boundaries and larger homogeneous misclassified regions. The proposed method explicitly models heterogeneous spectral and spatial characteristics in a decoupled manner, in contrast to Transformer-based models, which rely on global token interactions, and Mamba-based models, which focus on long-range propagation. This results in enhanced boundary preservation and class differentiation, as evidenced by the qualitative findings. The average feature map visualizations are shown in Fig. 12: the feature map without the SAB, the feature map without the CAB, and the feature map with both blocks included (full model). It is clear from

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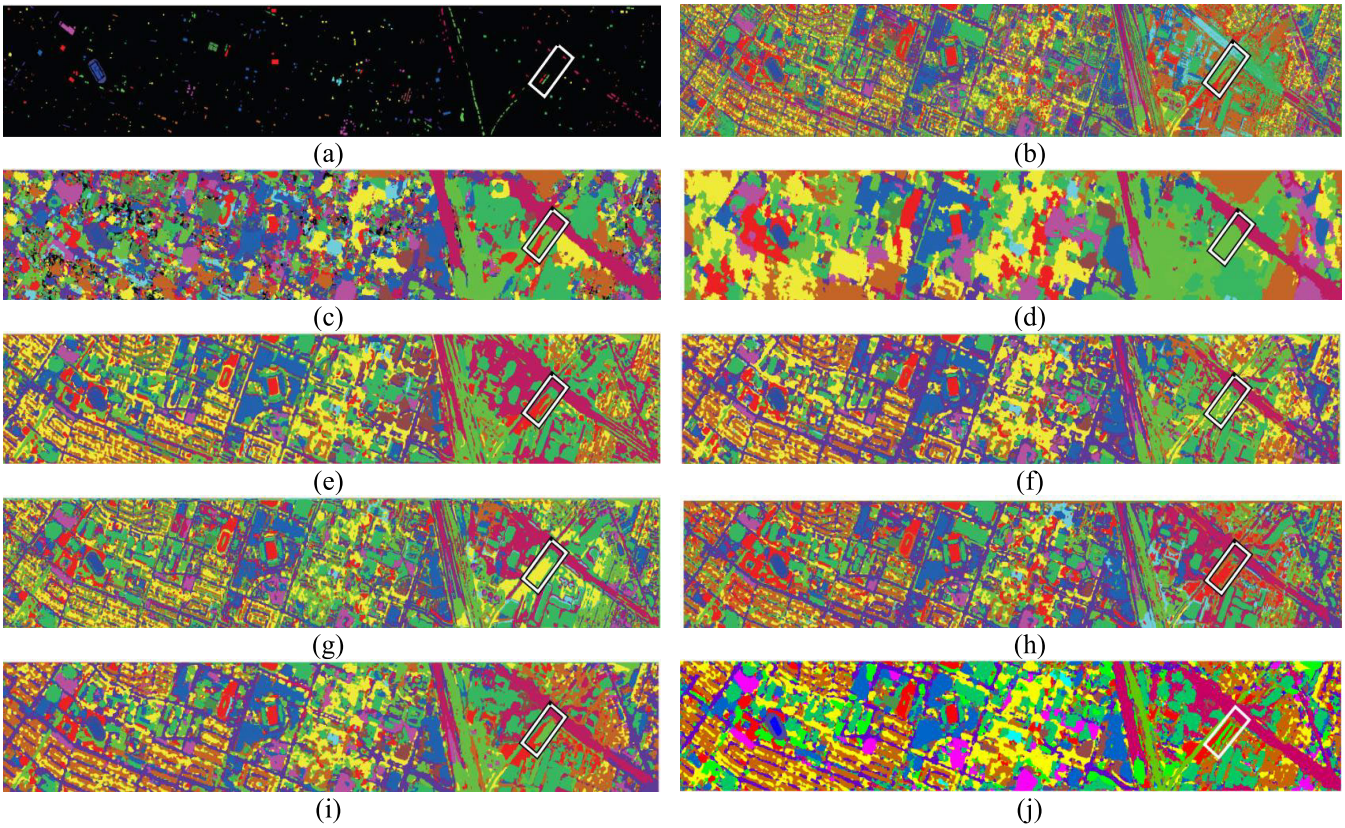


Fig. 9. Classification map of the Houston dataset is presented qualitatively. (a) Map of verified accuracy. (b) Support vector machine. (c) DMSGer. (d) GiGCN. (e) FullyContNet. (f) CLOLN. (g) Spectralformer. (h) GSC-ViT. (i) MambaHSI. (j) MogaHSI. Fig. 5(c) serves as the basis for color interpretation.

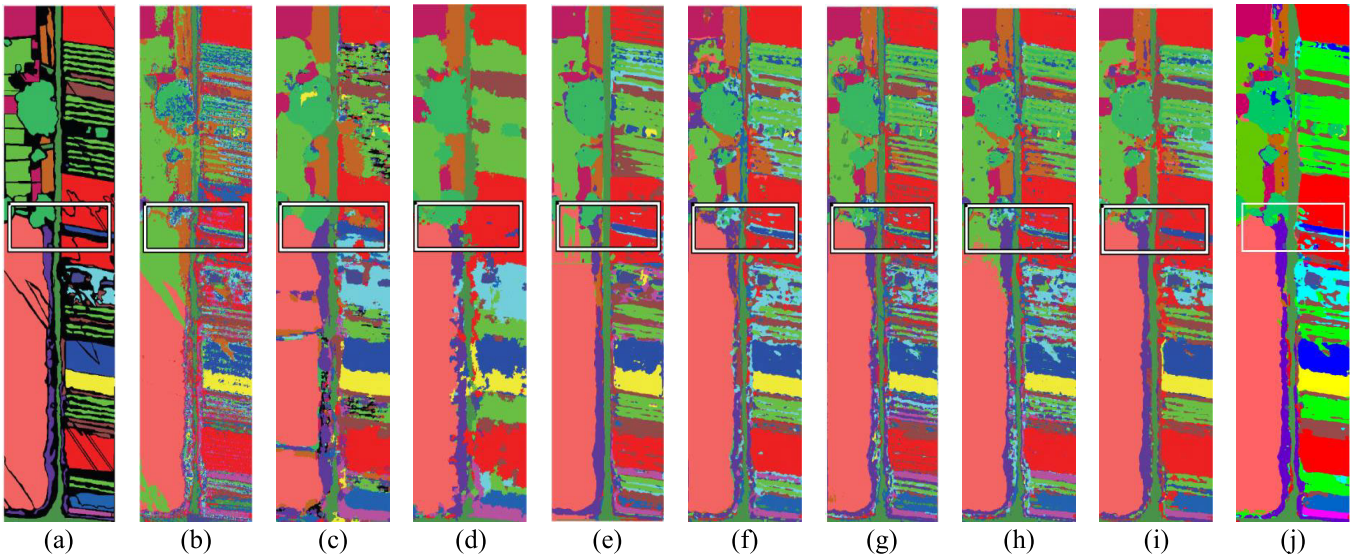


Fig. 10. Classification map of the HanChuan dataset is presented qualitatively. (a) Map of verified accuracy. (b) Support vector machine. (c) DMSGer. (d) GiGCN. (e) FullyContNet. (f) CLOLN. (g) Spectralformer. (h) GSC-ViT. (i) MambaHSI. (j) MogaHSI. Fig. 6(c) serves as the basis for color interpretation.

the figure that the block details are much clearer than in the other two cases. Also, the t-SNE visualization of the classes in the Pavia dataset is shown in Fig. 13. Each point represents a sample, colored by its class label, illustrating how class separability and clustering improve across the three configurations. All t-SNE plots were generated using scikit-learn with `n_components= 2`, `perplexity= 30`, `learning_rate= 200`, `n_iter= 1000`, `early_exaggeration= 12`, `metric='euclidean'`, `init='pca'`, and a fixed `random_state= 42`. Based on data in Fig. 4, it is clear that there is much improvement for the Trees/Meadows and Bricks/Baresoil when the CAB block is included because the CAB can make channel mixing. More compact intraclass features and clearer interclass separability result from CAB's adaptive aggregation of channel-wise responses, which highlights informative spectral bands while suppressing redundant or irrelevant ones.

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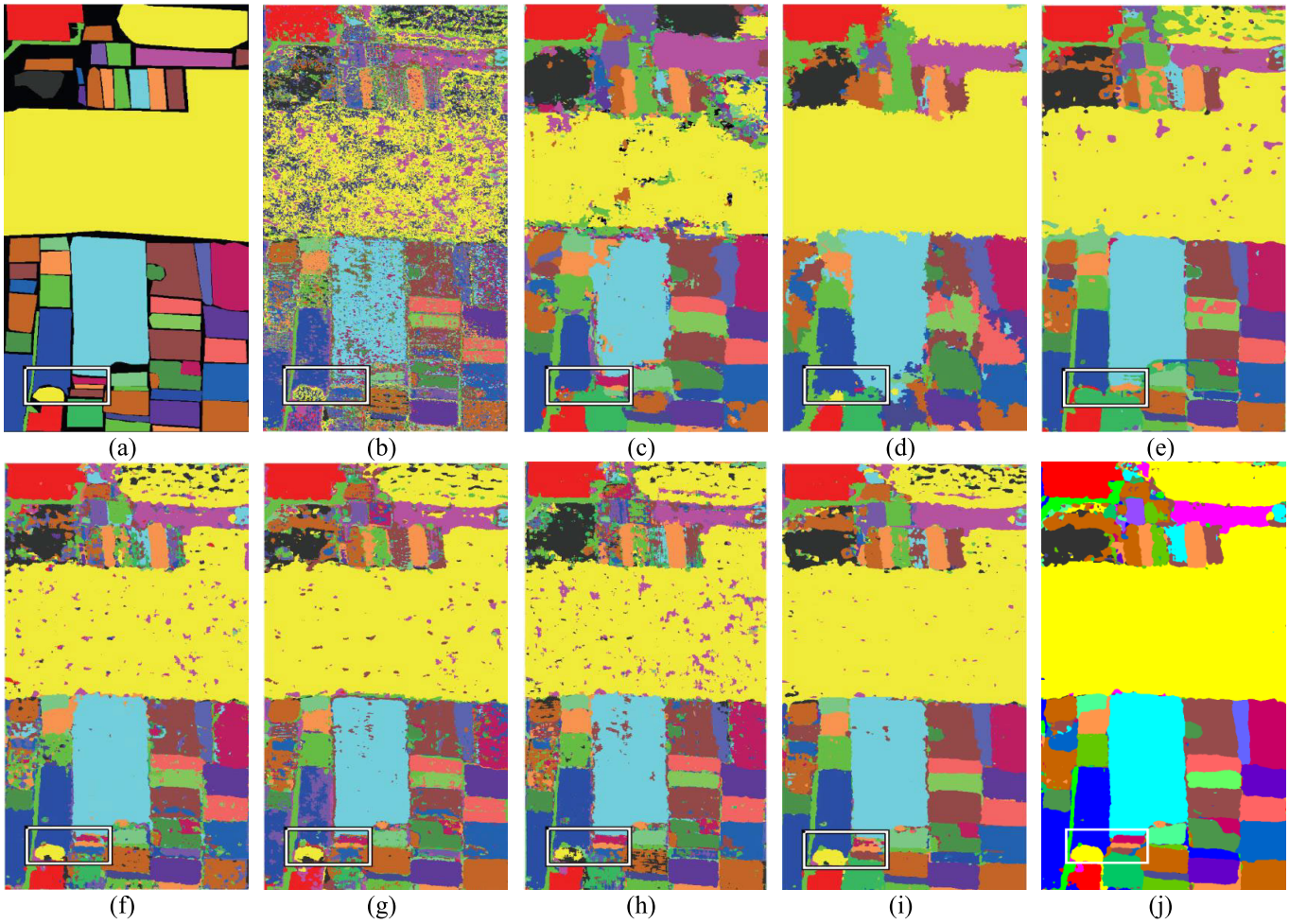


Fig. 11. Classification map of the HongHu dataset is presented qualitatively. (a) Map of verified accuracy. (b) Support vector machine. (c) DMSGer. (d) GiGCN. (e) FullyContNet. (f) CLOLN. (g) Spectralformer. (h) GSC-ViT. (i) MambaHSI. (j) MogaHSI. Fig. 7(c) serves as the basis for color interpretation.

D. Complexity Analysis

A detailed comparison of model parameters and floating-point operations (FLOPs) against representative SOTA methods are conducted on the Pavia dataset, which was summarized in Table VI. The proposed MogaHSI requires only 32% of the parameters demanded by MambaHSI, translating into a 39% reduction in computational cost. Furthermore, MogaHSI is 1.19% of GSC-ViT while delivering superior classification accuracy. These results collectively confirm that the proposed framework strikes a favorable balance between predictive performance and computational efficiency, making it well-suited for resource-constrained deployment scenarios.

E. Ablation Study

To evaluate each module's impact on computational cost and performance, we conducted an ablation study. The PaviaU dataset is the source of all parameters and FLOPS used in the ablation studies. We use HDmin values of 32, 64, and 128 to investigate the impact of the model's HDim. We then investigate the effect of model depth. The effect of adding a pooling layer next to every encoder block is then investigated. We also look at the effects of using SAB and CAB. We also showed that the HDim of the CAB block can be removed. Additionally, we used BN and GN to do an ablation. Additionally, we used attention force float point 32 (attn_force_fp32) to conduct ablation. Lastly, we performed an ablation of the mode's activation function. We conduct ablation studies on SAB and CAB to assess the effectiveness of heterogeneous spectral–spatial modeling. Removing SAB degrades performance, especially in scenes with complex spatial layouts. This shows that it is essential to model spatial heterogeneity. Excluding CAB also reduces accuracy, underscoring the importance of adaptive spectral reallocation for addressing spectral heterogeneity.

1) Impact of HDim: To illustrate the effect of model depth (HDim) on performance, we conducted an ablation study. Table VII displays the findings. Three models with HDim depths of 64, 128, and 32 are examined. According to the model's results, a depth of 64 can improve performance while balancing computational cost and performance. For example, the method with HDim = 64 has approximately 29% fewer parameters than the model with HDim = 128, and its performance improves slightly.

2) Impact on the Model Depth Dataset From PaviaU: Our goal in this ablation study was to show how model depth affects various datasets. The outcome of this ablation is seen in

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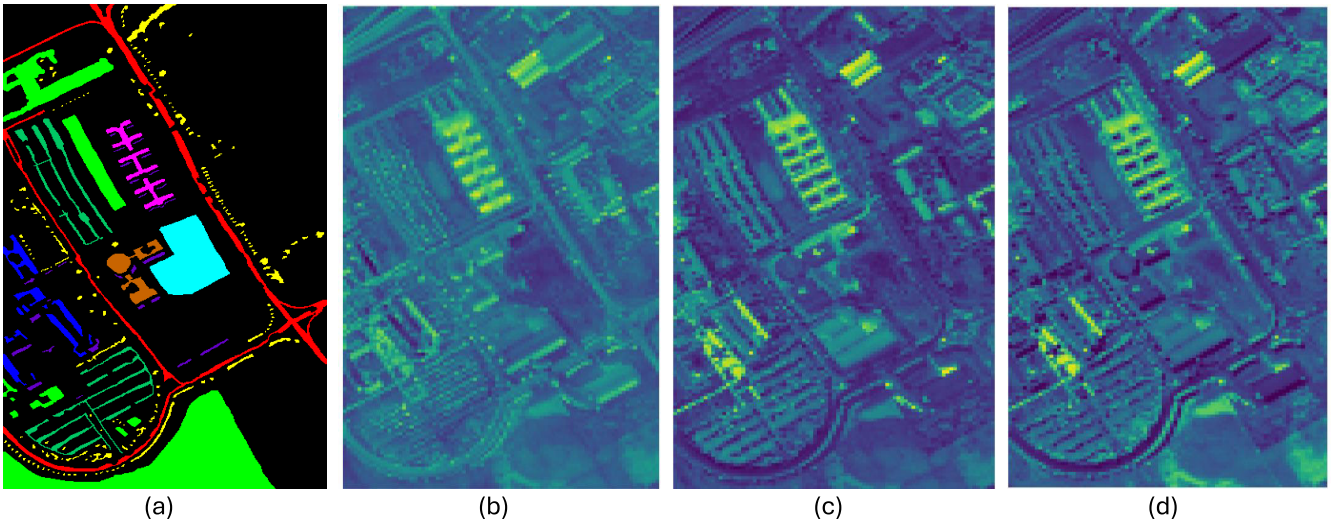


Fig. 12. Average feature map visualizations of all channels at the end layer of the encoder block. (a) Ground truth image, (b) model without CAB, (c) model without SAB, and (d) model with both blocks included.

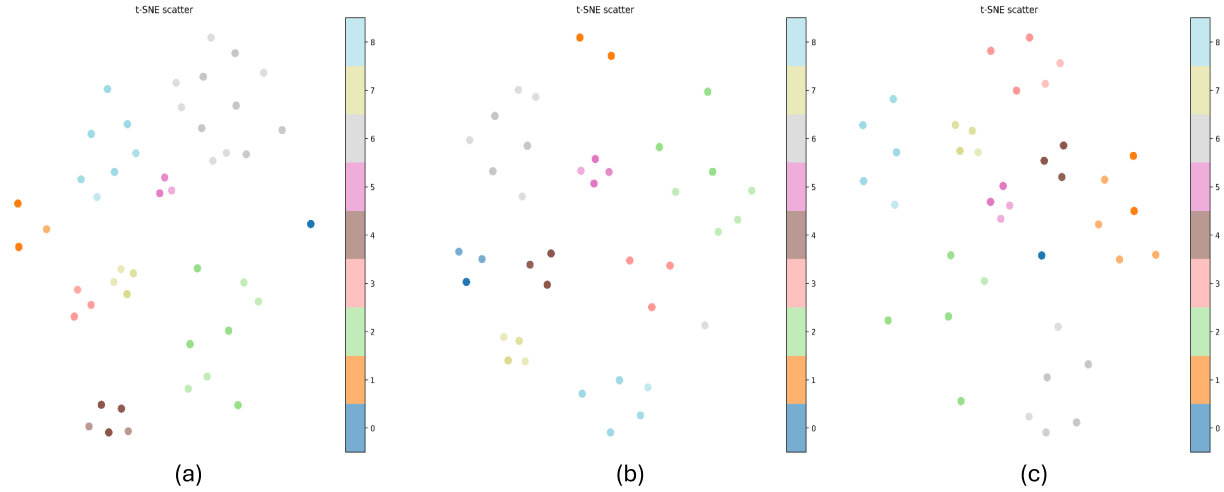


Fig. 13. Visualization of t-SNE of the classes of the Pavia dataset. (a) Model with both blocks, SAB and CAB. (b) Model without SAB. (c) Model without CAB.

Table VIII. We created two additional models, one with depth 2 and one with depth 4. Compared with the other models in the table, the proposed method with depth 3 performs better. For example, on the HanChuan dataset, the developed method with depth 3 achieves a Kappa of 91.73, whereas models with depths 2 and 4 achieve 88.13 and 89.6, respectively. The results demonstrate that the depth-3 model achieves an optimal tradeoff between computational overhead and classification results.

3) Impact of Average Pooling: We wanted to show the effect of using a pooling layer next to the encoder block during this ablation. Table IX presents the outcome, comparing the last row with the initial row. Compared with the model without a pooling layer, the findings demonstrate the pooling layer's significant influence. For instance, in the HongHu dataset, the OA decreased from 96.04 to 89.66. Furthermore, the number of FLOPs is significantly reduced by employing the pooling layer, dropping from 55.02 to 24.15 G.

4) Impact of Using SAB and CAB Block: We attempted to demonstrate how SAB and CAB affected the model's performance. Table X presents the findings. One of the models we created for our research is shown in Fig. 1. The other three models, lacking SAB and CAB, are shown in Fig. 2. The results demonstrate that the proposed method outperforms the others. For example, when the CAB was not used, Kappa decreased from 98.63 to 97.88. Furthermore, the results, particularly for the HongHu dataset, are significantly affected by the scenario in which SAB is applied first.

5) Effect of the Hidden Dim (H) of the CAB: We wanted to show the impact of the CAB's HDim (H) in this ablation. The ablation results are indicated in Table XI. Models with 1, 2, and 4 were compared. The chart indicates that the best-performing model has a HDim of 2. For the Pavia dataset, for instance, the model with a kernel size of 2 attained a Kappa of 98.63, which is higher than the other two cases. For most datasets, using H = 2 can improve the outcome.

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TABLE VI

COMPUTATIONAL COST EVALUATION ON THE PAVIAU DATASET. $FLOPs_{IMAGE}$ DENOTES THE TOTAL FLOPS REQUIRED TO PROCESS A COMPLETE HYPERSPECTRAL IMAGE. *M* AND *G* REFER TO MILLION AND BILLION, RESPECTIVELY. THE SUPERSCRIPTS † AND‡ INDICATES PATCH-LEVEL AND IMAGE-LEVEL METHODS, RESPECTIVELY

Metrics	**GCN-based**		**CNN-based**		**Transformer-based**		**SSM-based**	**SSM-based**
	DMSGer‡ *TNNLS2022*	GiGCN‡ *TNNLS2022*	FullyContNet‡ *TGRS2022*	CLOLN† *TGRS2024*	Spectralformer† *TGRS2021*	GSC-ViT† *TGRS2024*	MambaHSI‡ *TGRS2024*	MogaHSI‡ *Ours*
Parameters (M)	0.090	0.223	1.200	0.005	0.177	0.153	0.412	0.134
Memory (M)	-	-	593	130	556	364	5777	2258
$FLOPs_{image}$ (G)	7.66	11.01	145.62	124.44	9086.19	2026.298	39.18	24.15
OA(%)	89.55±1.22	94.47±0.97	91.31±9.36	94.97±1.43	91.10±2.60	92.65±2.30	95.74±0.90	**97.83±0.68**

TABLE VII

IMPACT OF HDIM SIZE ON PERFORMANCE, FLOPS, AND PARAMETERS

	Pavia			HanChun			HongHu			Houston				
	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	Param (K)	Flops (G)
ModelHDim=128	95.77±1.77	97.24±0.67	96.63±2.63	92.28±1.23	92.6±0.6	90.8±2.51	96.22±0.25	96.28±0.39	95.48±1.52	94.49±1.03	95.24±0.91	94.43±2.04	462.64	83.93
Proposed Model HDim=64	97.83±0.68	98.1±0.58	98.63±1.31	92.73±1.2	92.53±0.63	91.73±2.59	96.04±0.52	96.02±0.38	94.9±1.0	95.19±0.7	95.85±0.72	95.5±1.6	133.79	24.15
Model HDim=32	94.9±2.41	96.65±1.05	97.38±2.05	91.08±1.26	90.29±1.44	89.73±1.96	88.24±0.86	89.51±0.86	87.71±2.41	94.62±1.27	95.34±1.13	94.93±1.61	43.09	7.64

TABLE VIII

IMPACT OF MODEL DEPTH [NUMBER OF ENCODERS (NB) AND POOLING BLOCKS (PBS)]

	Pavia			HanChun			HongHu			Houston				
	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	Param (K)	Flops (G)
Proposed Model (NB=3, PB=2)	97.83±0.68	98.1±0.58	98.63±1.31	92.73±1.2	92.53±0.63	91.73±2.59	96.04±0.52	96.02±0.38	94.9±1.0	95.19±0.7	95.85±0.72	95.5±1.6	133.79	24.15
Model (NB=2, PB=1)	95.13±2.26	97.28±0.87	97.0±2.03	89.1±1.45	88.34±1.49	88.13±2.15	92.98±1.16	92.69±0.84	91.71±1.76	95.23±1.18	95.97±1.1	95.71±1.72	94.7	23.91
Model (NB=4, PB=3)	95.39±0.75	94.86±0.82	93.5±3.0	92.07±1.05	90.81±1.35	89.6±1.74	96.23±0.27	96.19±0.2	96.19±0.64	90.87±0.93	91.8±1.0	89.36±2.79	172.88	24.21

TABLE IX

IMPACT OF THE POOLING LAYER ON MODEL PERFORMANCE, PARAMETERS, AND FLOPS

	Pavia			HanChun			HongHu			Houston				
	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	Param (K)	Flops (G)
Proposed Model	97.83±0.68	98.1±0.58	98.63±1.31	92.73±1.2	92.53±0.63	91.73±2.59	96.04±0.52	96.02±0.38	94.9±1.0	95.19±0.7	95.85±0.72	95.5±1.6	133.79	24.15
Model W/O Pooling	94.46±1.6	96.09±1.01	96.0±2.22	82.66±2.16	81.45±1.78	78.2±2.7	89.66±1.31	88.9±0.75	87.05±1.65	95.22±1.22	95.93±1.04	95.86±2.16	133.79	55.02

TABLE X

IMPACT OF THE SAB AND CAB ON MODEL PERFORMANCE, PARAMETERS, AND FLOPS

	Pavia			HanChun			HongHu			Houston				
	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	Param (K)	Flops (G)
Proposed Model	97.83±0.68	98.1±0.58	98.63±1.31	92.73±1.2	92.53±0.63	91.73±2.59	96.04±0.52	96.02±0.38	94.9±1.0	95.19±0.7	95.85±0.72	95.5±1.6	133.79	24.15
Model W/O CAB	96.82±1.35	97.7±0.62	97.88±2.17	93.35±1.27	92.7±0.59	92.87±2.23	95.32±0.71	95.19±0.48	94.33±1.27	94.77±0.78	95.4±0.75	95.0±2.09	79.45	14.33
Model W/O SAB	95.6±1.98	96.72±1.17	97.13±2.24	88.24±1.75	86.88±1.79	85.6±3.13	90.98±0.84	90.86±0.66	90.05±2.06	95.27±0.51	95.82±0.49	95.29±1.32	72.01	12.92

6) Effects of BN and Group Norm: We conducted a study on the batch norm (BN) and GN during this last ablation. Table XII presents the findings. The table clearly indicates that the model with GN can achieve its peak performance. For example, the OA of the HongHu dataset decreased from 96.04 to 95.56.

7) Effects of Attention Force Float Point 32 (Attn_Force_Fp32): We conducted a study on the att_force_fp32 during this last ablation. Table XIII presents the findings. The table indicates that the model without this function can achieve its peak performance. For example, the OA of the HanChuan dataset improved by 0.79.

8) Activation Function Impact: We designed a study on the GELU, ReLU, and SiLU activations during this ablation. Table XIV presents the findings. The table indicates that the model with GELU performs better on most datasets. For example, the OA of the Pavia dataset decreased from 97.83 to 97.54.

9) Ablation Study on TV Loss: We performed an ablation study on total variation (TV) as spectral–spatial regularization. We weighted the TV with weight λ. The weight λ is set to four values, ranging from 0.01 to 10. The result is indicated in Table XV. It is clear that when the TV is used, its performance is badly affected. TV regularization can hurt performance in HSI classification because its core assumption—that neighboring pixels should be similar—often conflicts with the intrinsic properties of hyperspectral data and modern deep models.

10) Ablation Study on the Impact of Using Self-Attention: We did an ablation study on self-attention mechanisms. We attempted to use the conventional self-attention mechanism for this ablation study, but because the model operates on whole images, it exceeded our GPU memory capacity. To address this issue, we used a variant of self-attention, namely self-attention with bidirectional interaction, proposed in [55] and [56]. Additionally, this model did not fit within our GPU memory, so we reduced the feature size from 64 to 8 in the proposed model. The result is shown in Table XVI. The proposed model shows better performance across many datasets. We also note that self-attention requires more training time, so we include this in our comparison.

11) Effects of the Moga Block: We designed a study on the Moga impact during this ablation. Table XVII presents

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TABLE XI

IMPACT OF HIDDEN DIM (H) OF THE CAB

	Pavia			HanChun			HongHu			Houston				
	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	Param (K)	Flops (G)
Proposed Model CAB H=2	97.83±0.68	98.1±0.58	98.63±1.31	92.73±1.2	92.53±0.63	91.73±2.59	96.04±0.52	96.02±0.38	94.9±1.0	95.19±0.7	95.85±0.72	95.5±1.6	133.79	24.15
Model CAB H=1	97.13±0.93	97.67±0.79	97.88±2.24	92.94±1.23	92.16±1.26	91.27±2.88	95.9±0.44	95.74±0.43	94.48±1.25	94.73±1.54	95.46±1.37	95.29±1.47	106.72	19.29
Model CAB H=4	97.15±1.25	97.91±0.88	97.62±2.53	92.81±1.54	92.78±1.0	92.13±1.57	96.2±0.38	96.04±0.35	95.57±1.83	95.24±0.85	95.89±0.85	95.21±1.01	187.93	33.77

TABLE XII

EFFECTS OF BN AND GROUP NORM

	Pavia			HanChun			HongHu			Houston				
	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	Param (K)	Flops (G)
Proposed Model GN	97.83±0.68	98.1±0.58	98.63±1.31	92.73±1.2	92.53±0.63	91.73±2.59	96.04±0.52	96.02±0.38	94.9±1.0	95.19±0.7	95.85±0.72	95.5±1.6	133.79	24.15
Model BN	97.7±0.88	98.08±0.7	98.13±1.96	92.52±1.39	92.48±0.64	91.87±2.86	95.56±0.98	95.49±0.6	94.57±0.96	95.15±0.86	95.84±0.82	95.5±1.63	133.79	24.05

TABLE XIII

EFFECTS OF ATTENTION FORCE FLOAT POINT 32 (ATTN_FORCE_FP32) ON THE PERFORMANCE

	Pavia			HanChun			HongHu			Houston				
	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	Param (K)	Flops (G)
Proposed Model	97.83±0.68	98.1±0.58	98.63±1.31	92.73±1.2	92.53±0.63	91.73±2.59	96.04±0.52	96.02±0.38	94.9±1.0	95.19±0.7	95.85±0.72	95.5±1.6	133.79	24.15
Model attn_force_fp32	97.49±1.21	97.96±0.83	98.38±1.59	91.94±1.79	91.76±0.79	91.0±2.13	96.01±0.5	96.07±0.42	95.1±1.4	94.92±0.69	95.64±0.68	95.36±1.61	133.79	24.05

TABLE XIV

EFFECTS OF THE ACTIVATION FUNCTION ON THE PERFORMANCE

	Pavia			HanChun			HongHu			Houston				
	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	QA	AA	Kappa	Param (K)	Flops (G)
Proposed Model GELU	97.83±0.68	98.1±0.58	98.63±1.31	92.73±1.2	92.53±0.63	91.73±2.59	96.04±0.52	96.02±0.38	94.9±1.0	95.19±0.7	95.85±0.72	95.5±1.6	133.79	24.15
Model ReLU	97.54±1.29	98.0±0.71	98.38±1.59	93.3±1.04	92.93±0.94	92.27±2.35	95.89±0.5	95.73±0.41	94.9±1.19	95.22±0.71	95.94±0.66	95.57±1.46	133.79	24.05
Model SiLU	97.59±0.8	97.92±0.66	98.25±1.6	92.86±1.25	92.59±1.05	92.27±2.43	95.74±0.59	95.51±0.55	94.38±0.9	94.96±0.84	95.67±0.87	95.21±1.94	133.79	24.05

TABLE XV

EFFECTS OF THE TV LOSS ON THE PERFORMANCE

	Pavia			HanChun			HongHu			Houston				
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	Param (K)	Flops (G)
Proposed Model	97.83±0.68	98.1±0.58	98.63±1.31	92.73±1.2	92.53±0.63	91.73±2.59	96.04±0.52	96.02±0.38	94.9±1.0	95.19±0.7	95.85±0.72	95.5±1.6	133.79	24.15
Model $\lambda = 0.01$	97.84±0.68	98.11±0.57	98.38±1.37	92.68±1.13	92.55±0.64	91.87±2.66	96.05±0.51	96.02±0.37	95.05±0.96	95.21±0.77	95.89±0.77	95.64±1.51	133.79	24.15
Model $\lambda = 0.1$	97.84±0.55	98.03±0.44	98.5±1.46	93.18±1.29	92.89±0.7	92.6±2.76	96.64±0.3	96.41±0.26	95.81±0.7	94.8±0.88	95.53±0.84	94.36±2.4	133.79	24.15
Model $\lambda = 1$	97.08±0.76	97.09±0.63	95.75±2.18	94.24±1.65	93.47±1.12	94.0±1.91	96.64±0.3	96.41±0.26	95.81±0.7	93.66±1.1	94.61±1.03	89.57±4.54	133.79	24.15
Model $\lambda = 10$	85.09±3.15	88.98±2.45	88.5±2.49	93.18±1.93	92.04±1.7	91.07±2.27	95.24±0.53	94.8±0.48	93.57±1.21	84.42±1.84	86.65±1.8	80.57±3.5	133.79	24.15

TABLE XVI

EFFECTS OF USING A SELF-ATTENTION LAYER ON THE PERFORMANCE

	Pavia			HanChun			HongHu			Complexity		
Method	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	Time	Param (K)	FLOPs (G)
Proposed model	75.04 ± 4.07	79.56 ± 2.57	76.62 ± 5.09	71.91 ± 2.46	69.61 ± 2.68	68.20 ± 3.13	68.76 ± 4.22	68.24 ± 2.86	63.57 ± 3.42	12.5	7.33	1.10
Model w Self-attention	78.40 ± 7.23	86.07 ± 2.04	86.12 ± 3.38	68.55 ± 5.05	65.53 ± 2.96	63.20 ± 4.73	70.53 ± 7.06	60.68 ± 4.52	58.24 ± 5.34	78.75	7.53	0.70

TABLE XVII

EFFECTS OF THE MOGA LAYER ON THE PERFORMANCE

	Pavia			HanChun			HongHu			Houston				
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	Param (K)	Flops (G)
Proposed Model	97.83±0.68	98.1±0.58	98.63±1.31	92.73±1.2	92.53±0.63	91.73±2.59	96.04±0.52	96.02±0.38	94.9±1.0	95.19±0.7	95.85±0.72	95.5±1.6	133.79	24.15
Model W/O Moga	96.16±1.48	97.16±0.67	97.25±1.66	87.75±2.15	86.19±1.47	85.73±2.25	91.2±0.72	90.87±0.59	90.33±1.24	995.44±0.49	96.06±0.46	95.86±1.65	109.6	19.8

TABLE XVIII

PER-CLASS AC (ACC% ± STD%) ON THE PAVIAU DATASET WITH AND WITHOUT THE CAB

Model	Asphalt	Meadows	Gravel	Trees	Metal sheets	Bare soil	Bitumen	Bricks	Shadows	Avg (Mean)
Model W/O CAB	96.40 ± 1.72	96.12 ± 3.03	98.39 ± 3.43	94.03 ± 1.97	99.93 ± 0.14	99.32 ± 0.74	99.86 ± 0.27	96.37 ± 2.20	98.85 ± 1.08	97.70
Our Model	95.29 ± 3.07	98.36 ± 1.63	97.93 ± 3.09	95.33 ± 1.89	99.95 ± 0.14	99.36 ± 0.82	99.81 ± 0.27	97.91 ± 1.03	98.93 ± 0.81	98.10

the findings. The table indicates that the model without Moga has experienced a performance drop across most datasets. For example, the OA of the HanChuan dataset decreased from 92.73 to 87.75. This indicates the effectiveness of the Moga in extracting multiorder gate aggregation.

12) Effects of the CAB on Different Classes of the Pavia Dataset: We designed a study on the CAB of different classes. Table XVIII presents the findings. The class names are shown in Fig. 4. The table indicates that the model without CAB has experienced a performance drop across most classes. For

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example, the OA of Meadows decreased from 98.36 to 96.12. This indicates the CAB's effectiveness in extracting multiorder gate aggregation.

V. Conclusion

In this work, we introduced MogaHSI, a lightweight HSI classification framework that explicitly models multiorder spatial-channel interactions to enhance spectral–spatial representation learning. MogaHSI's performance improvements come from three main things: 1) a compact multilevel interaction modeling framework that makes features more expressive while keeping the computational complexity low; 2) a good balance between pixel-level dense prediction and global contextual modeling, which keeps fine-grained spatial details while capturing long-range spectral–spatial dependencies; and 3) strong superiority and generalization ability across multiscene hyperspectral datasets, as shown by consistent improvements over ConvNet-, Transformer-, and Mamba-based approaches in both accuracy and efficiency. These findings validate the significance of explicitly modeling multiorder interactions for hyperspectral representation learning.

Future work will extend MogaHSI in three directions rooted in its core design. First, we will generalize the Moga mechanism to dense prediction tasks such as hyperspectral semantic segmentation and change detection, where multiscale boundary modeling is critical. Second, we will extend MogaHSI into a cross-modal fusion framework that integrates LiDAR-derived structural cues with HSI spectral features, using gated aggregation blocks to adaptively weight modality contributions—thereby directly addressing spectral confusion between structurally distinct materials. Third, we will explore adaptive interaction strategies to further improve robustness to irregular spatial distributions, overcoming the structural rigidity observed in current convolutional and sequential paradigms.

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